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NATIONAL SPACE ACADEMY

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Introduction to Hyperspectral Imaging

Light, spectra, and what we can measure with them
Space Academy 2026

What this session covers

contents

#	Section	What you will get out of it
01	Spectroscopy	How light is split into bands, and how a sensor measures them
02	Spectral signatures	Why every material has a fingerprint, and what a data cube is
03	Data acquisition	Field, drone, aircraft and satellite, and how the image is built
04	Processing HSI data	Calibration, binning, atmosphere and geometry
05	Challenges	Light, weather, movement and data size
06	Hyperspectral sensors	What is flying today and what is coming
07	Applications	Environment, industry, geology and gas detection

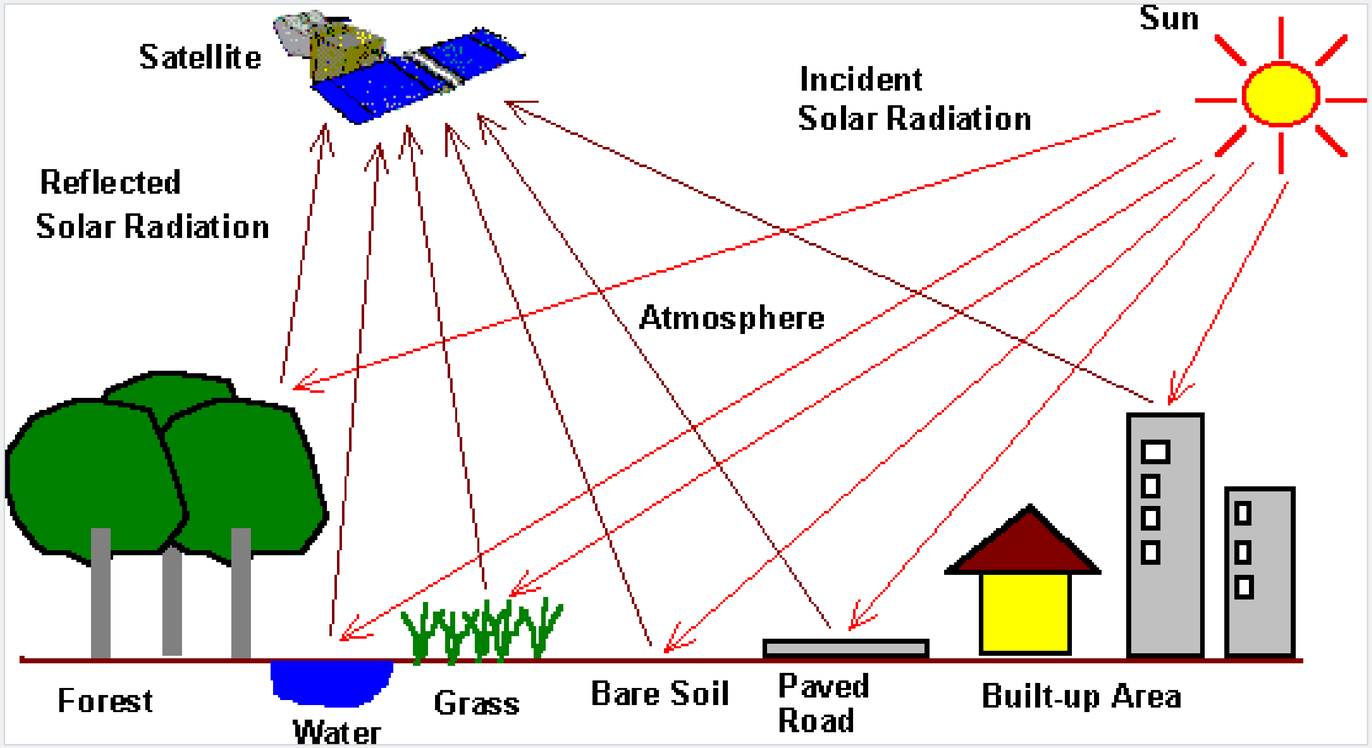
Hyperspectral imaging measures hundreds of narrow colour bands for every pixel, so we can tell materials apart, not just look at them.

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Recap · what remote sensing is

starting point

- **Measuring from a distance.** No contact with the target, so we can cover large areas quickly.
- **Always needs light.** Passive sensors use the sun. Active sensors, such as LiDAR and radar, bring their own.
- **Every object reflects differently.** That difference is what we measure and turn into a map.
- **Used everywhere.** Agriculture, forestry, defence, weather, mining and industry.



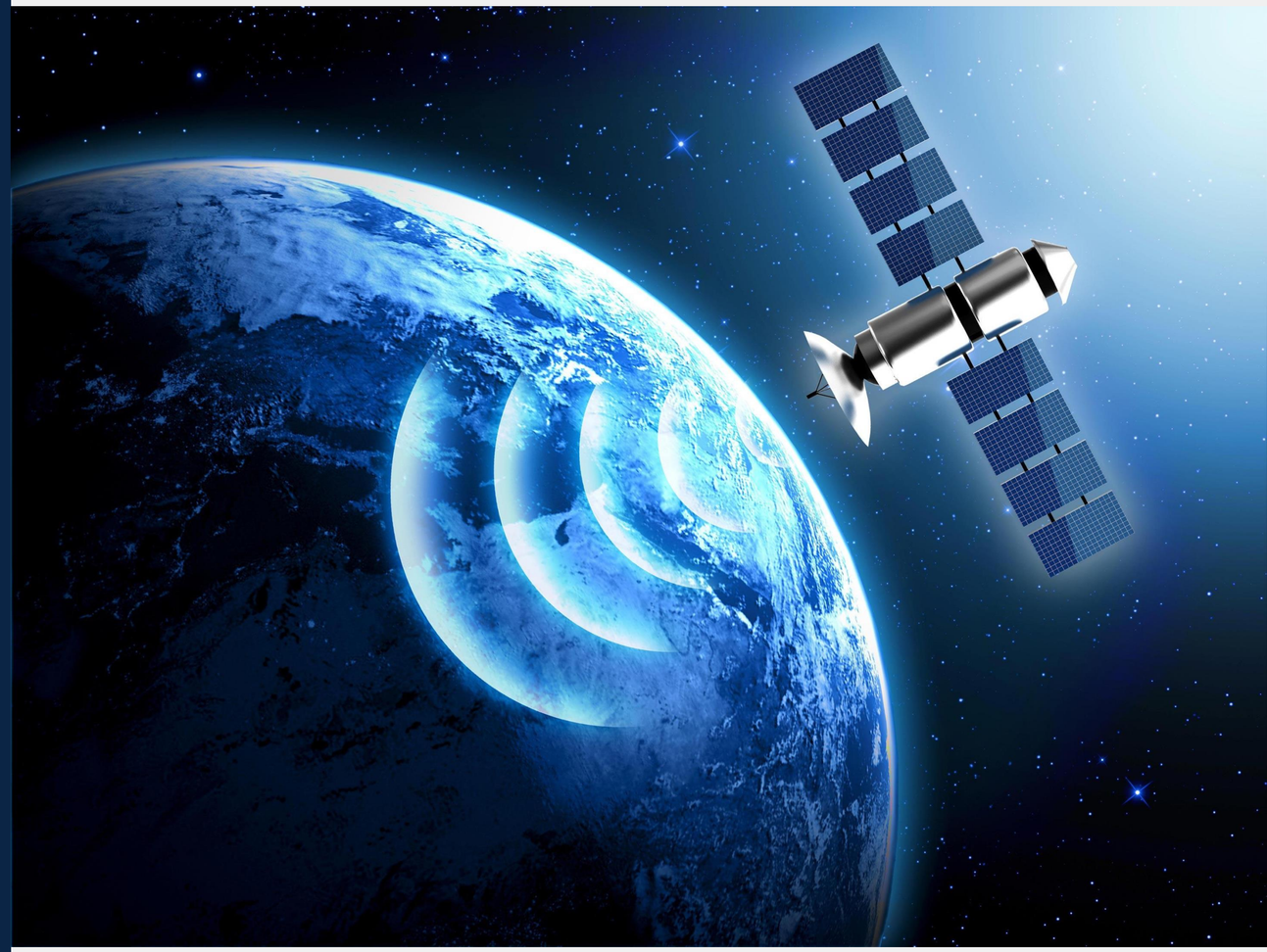
Sunlight reaches the surface, and part of it is reflected back to the sensor

Figure from the original UASA hyperspectral imaging course material

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01

Spectroscopy

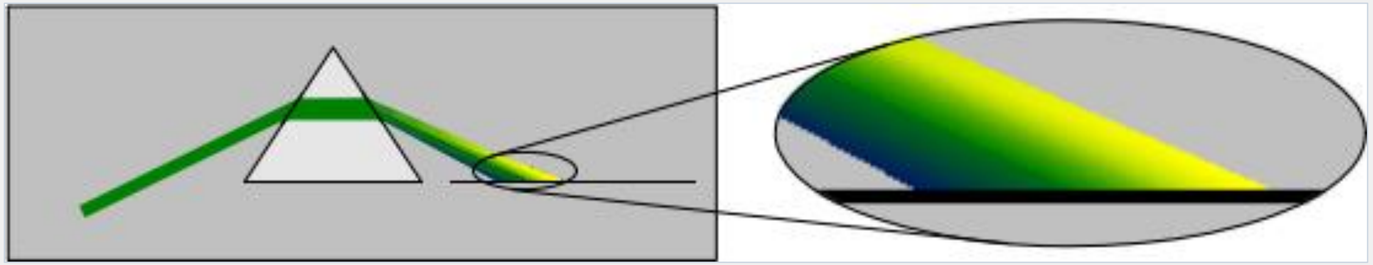


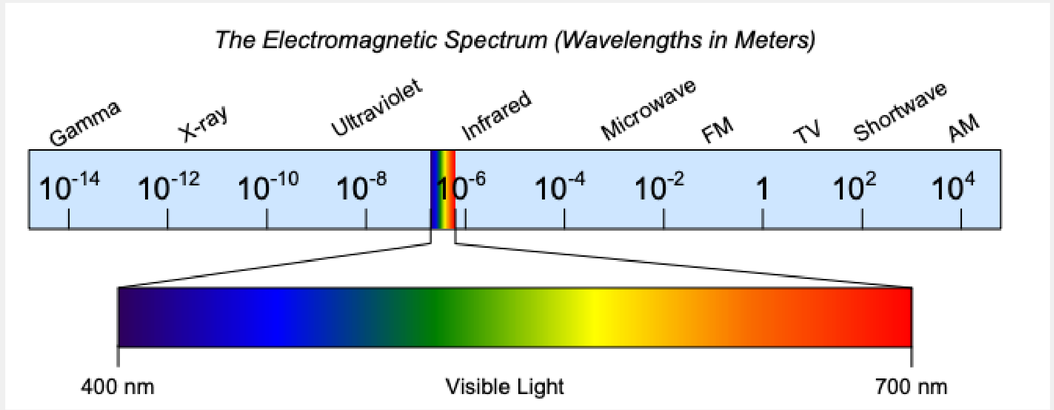
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Light, and how it splits

spectroscopy

- **Spectroscopy studies what light is made of:** It splits radiation by wavelength.
- **White light is a mixture.** A prism or a grating spreads it into its colours.
- **Light interacts with matter.** It can be transmitted, absorbed, reflected or refracted.
- **Our eyes stop at red.** Sensors continue into the near and shortwave infrared, where materials differ most.





White light spread into a continuous spectrum

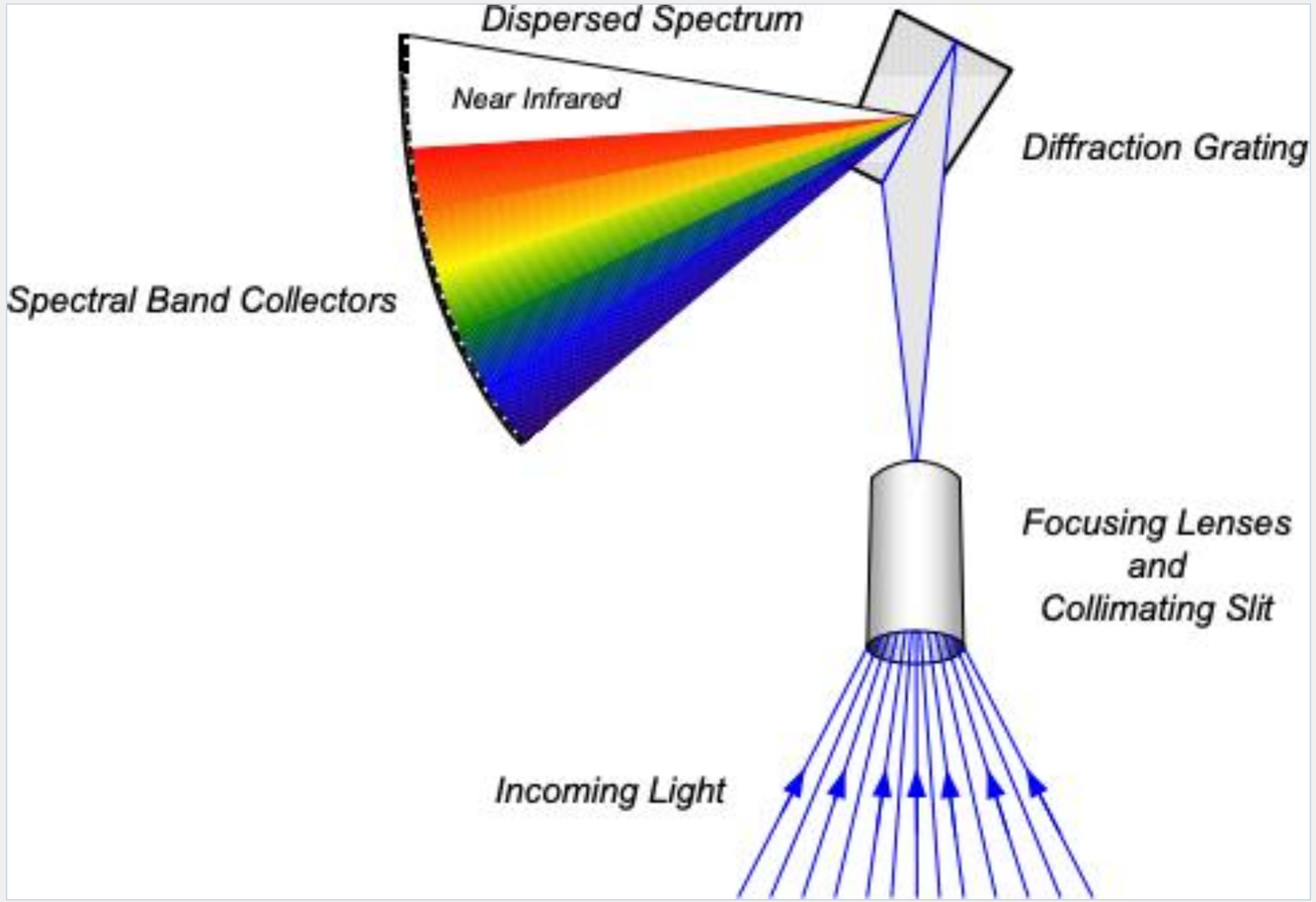
Figure from the original UASA hyperspectral imaging course material

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How a spectrometer works

the instrument

- **A slit lets a thin line of light in.** Lenses focus it onto the dispersing element.
- **A diffraction grating splits the light.** Fine etched lines send each wavelength in its own direction.
- **Detectors sit at the end.** Each one collects a narrow band and records its intensity.
- **Hundreds of bands per pixel.** That is what makes the data hyperspectral rather than multispectral.



Incoming light is focused, dispersed by a grating and collected band by band

Figure from the original UASA hyperspectral imaging course material

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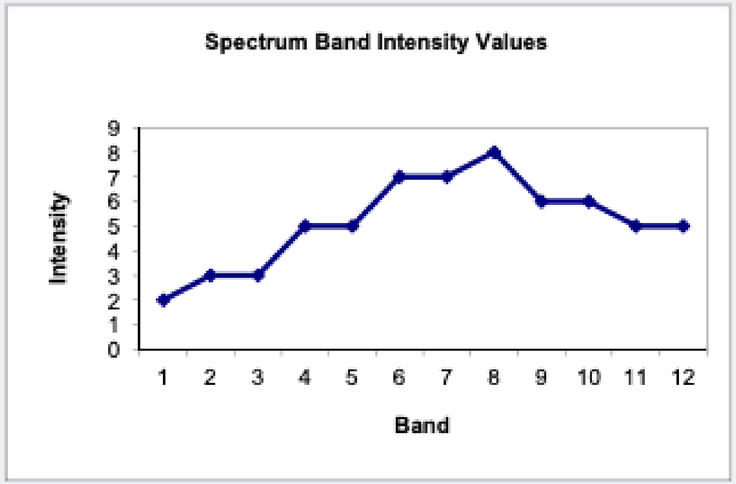
From light to numbers

`measurement`

- **Each band is measured separately.** The detector records an intensity for every one of them.
- **The list of values is the spectrum.** Plotted across the bands, those numbers become a curve.
- **The curve is the measurement.** It can be stored, compared and matched against reference spectra.
- **Same idea at every pixel.** A hyperspectral image repeats this measurement across the whole scene.

Band	Intensity Value	Band	Intensity Value
1	2	7	7
2	3	8	8
3	3	9	6
4	5	10	6
5	5	11	5
6	7	12	5

Each band gets an intensity value



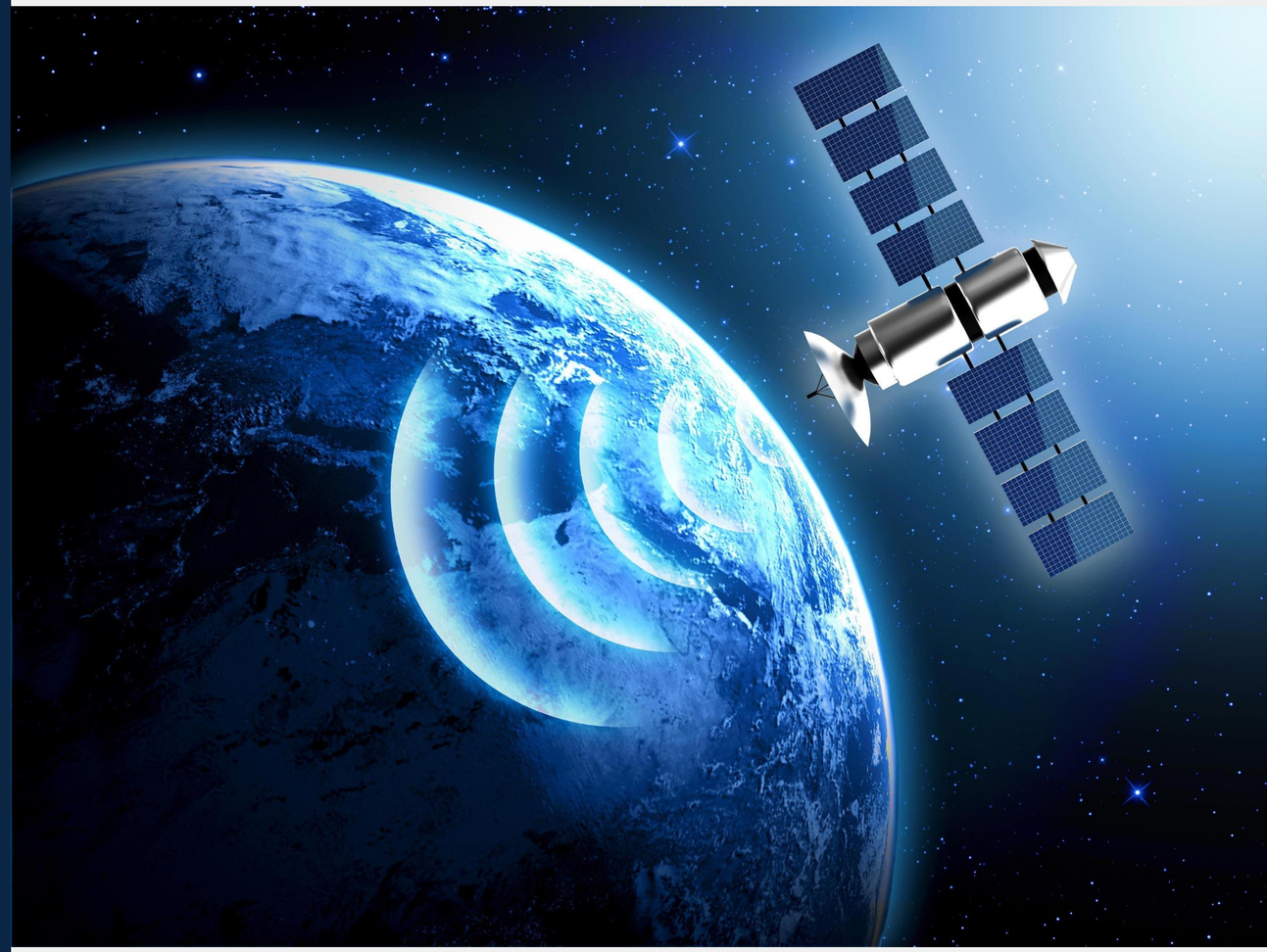
Plotted across the bands, the values become a curve

Figure from the original UASA hyperspectral imaging course material

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02

Spectral signatures

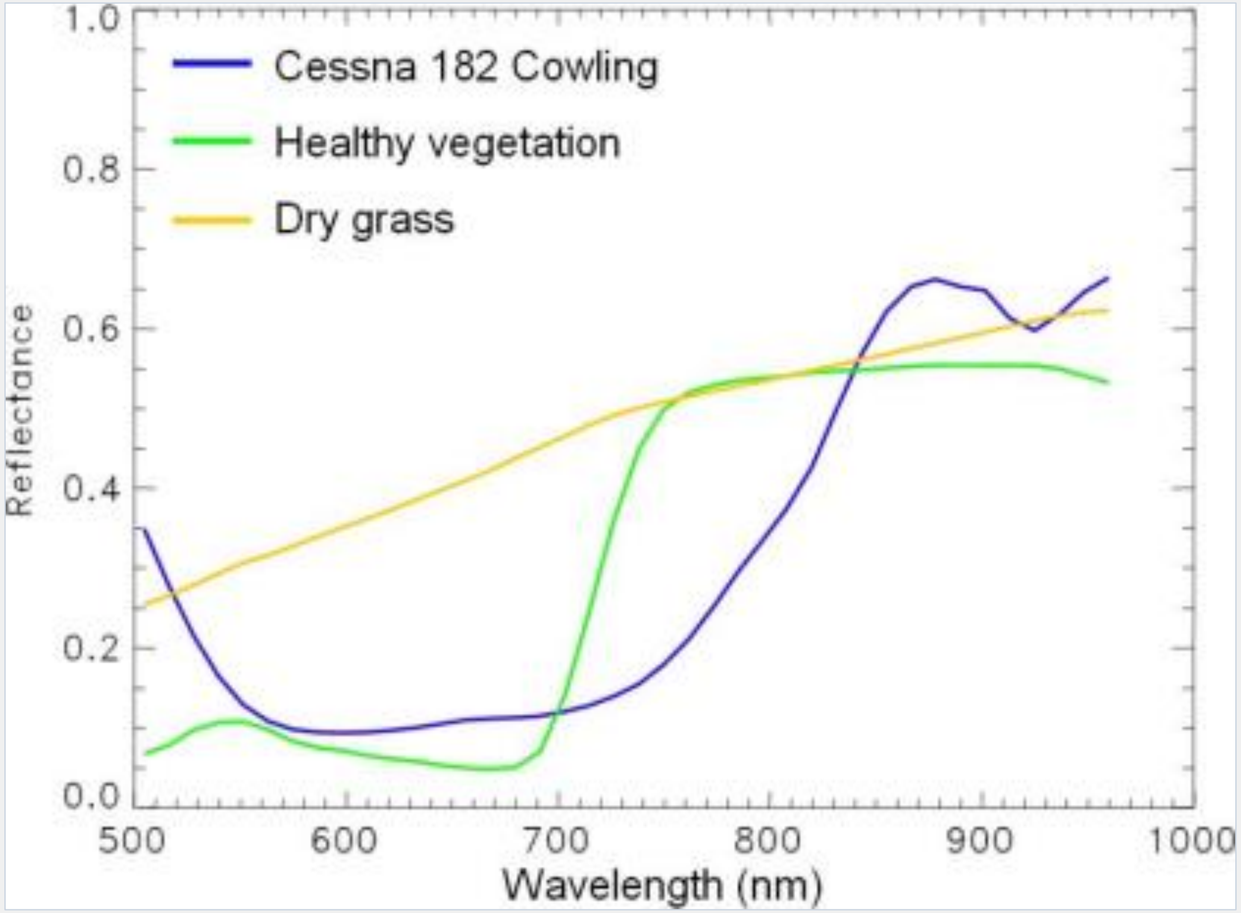


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Every material has a fingerprint

spectral signature

- **Reflectance is a physical property.** The share of incoming light returned at each wavelength.
- **The shape is the signature.** Healthy vegetation, dry grass and painted metal all look different.
- **Look at the jump near 700 nm.** Living plants reflect near-infrared strongly, which is the red edge.
- **This is why HSI works.** Objects that look alike to the eye separate clearly in the spectrum.



Three very different signatures measured over the same wavelengths

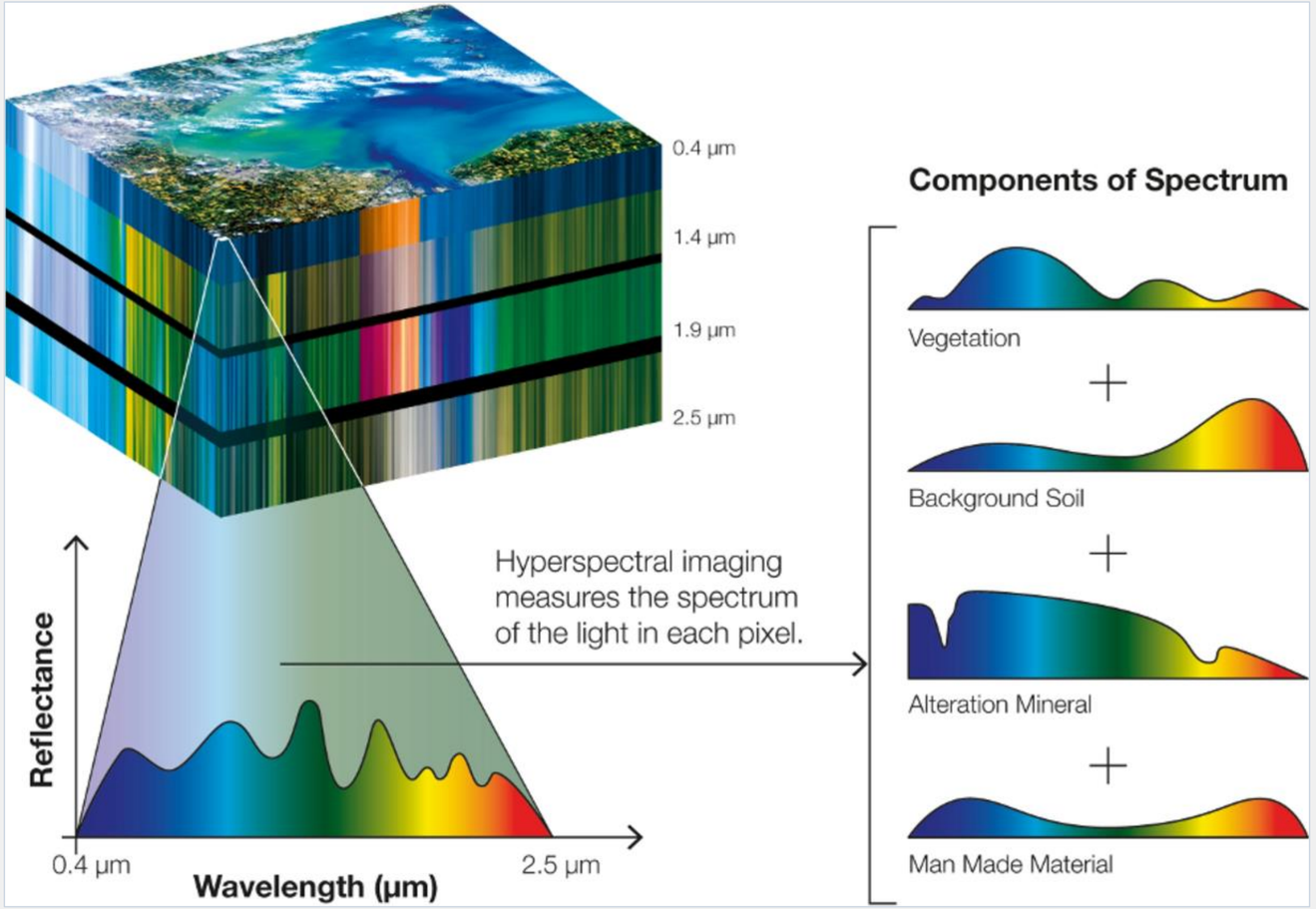
Figure from the original UASA hyperspectral imaging course material

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What hyperspectral imaging is

the data cube

- **Many narrow, touching bands.** Typically a hundred or more, covering the visible to shortwave infrared.
- **Two spatial dimensions, one spectral.** The result is a cube, with a full spectrum behind every pixel.
- **Each pixel can be identified.** Vegetation, soil, minerals and man-made materials have distinct curves.
- **The price is size.** One scene can be hundreds of bands deep, so storage and processing grow quickly.



A hyperspectral cube, and the spectra hidden inside it

Figure from the original UASA hyperspectral imaging course material

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Multispectral or hyperspectral?

how many bands

	Multispectral	Hyperspectral
Number of bands	about 4 to 12	often 100 to 300
Band width	wide, with gaps between them	narrow and contiguous
What you can tell apart	broad classes such as water, plants, built-up	materials and their condition
Typical platforms	many satellites, global coverage	fewer satellites, aircraft and drones
Revisit	high, every few days	lower, often by campaign
Data size	manageable	large and complex

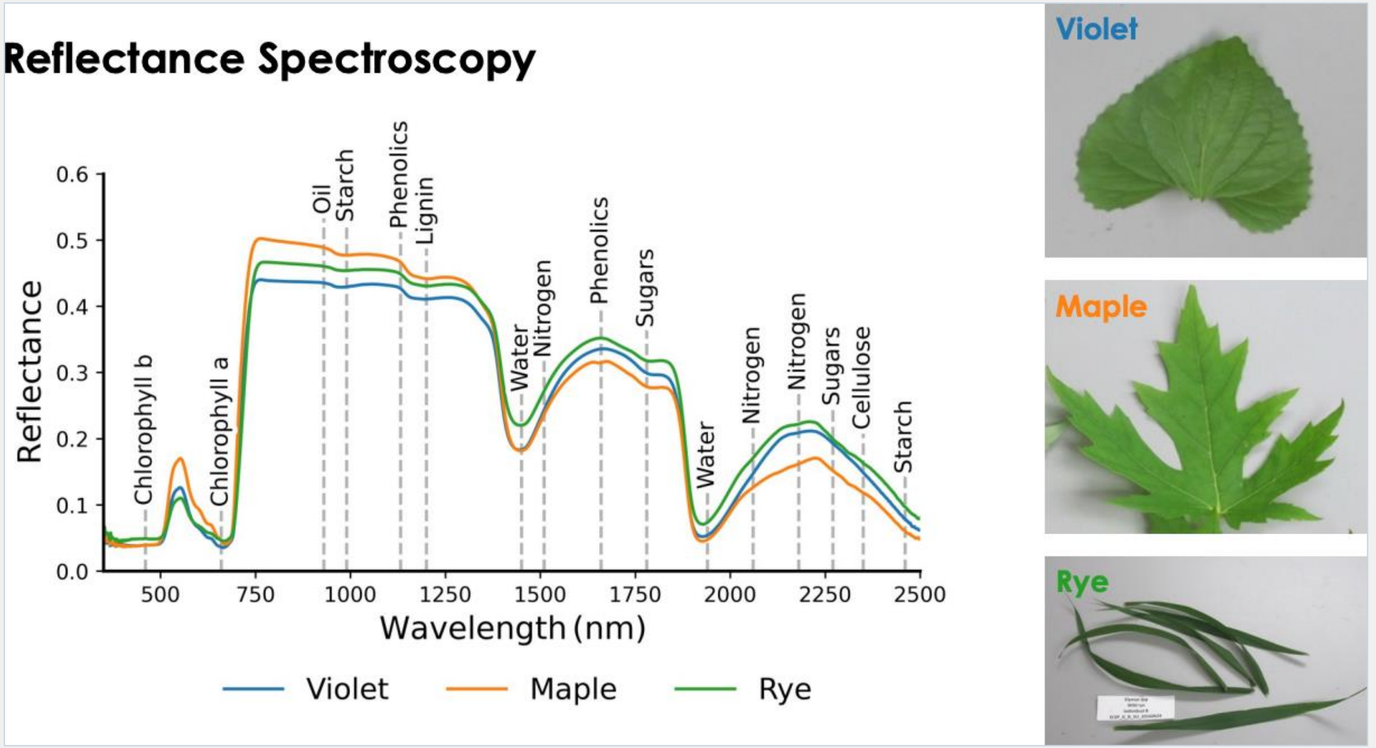
More bands mean better separability. Use multispectral to see where something is, and hyperspectral to work out what it is made of.

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Reading a spectrum

what the dips mean

- **Dips are absorption features.** Water, chlorophyll, nitrogen, sugars and cellulose each absorb at known wavelengths.
- **Position matters more than depth.** Where the dip sits tells you which material caused it.
- **Libraries make it practical.** Measured spectra are matched against reference libraries.
- **Narrow bands are essential.** Wide multispectral bands average these features away.



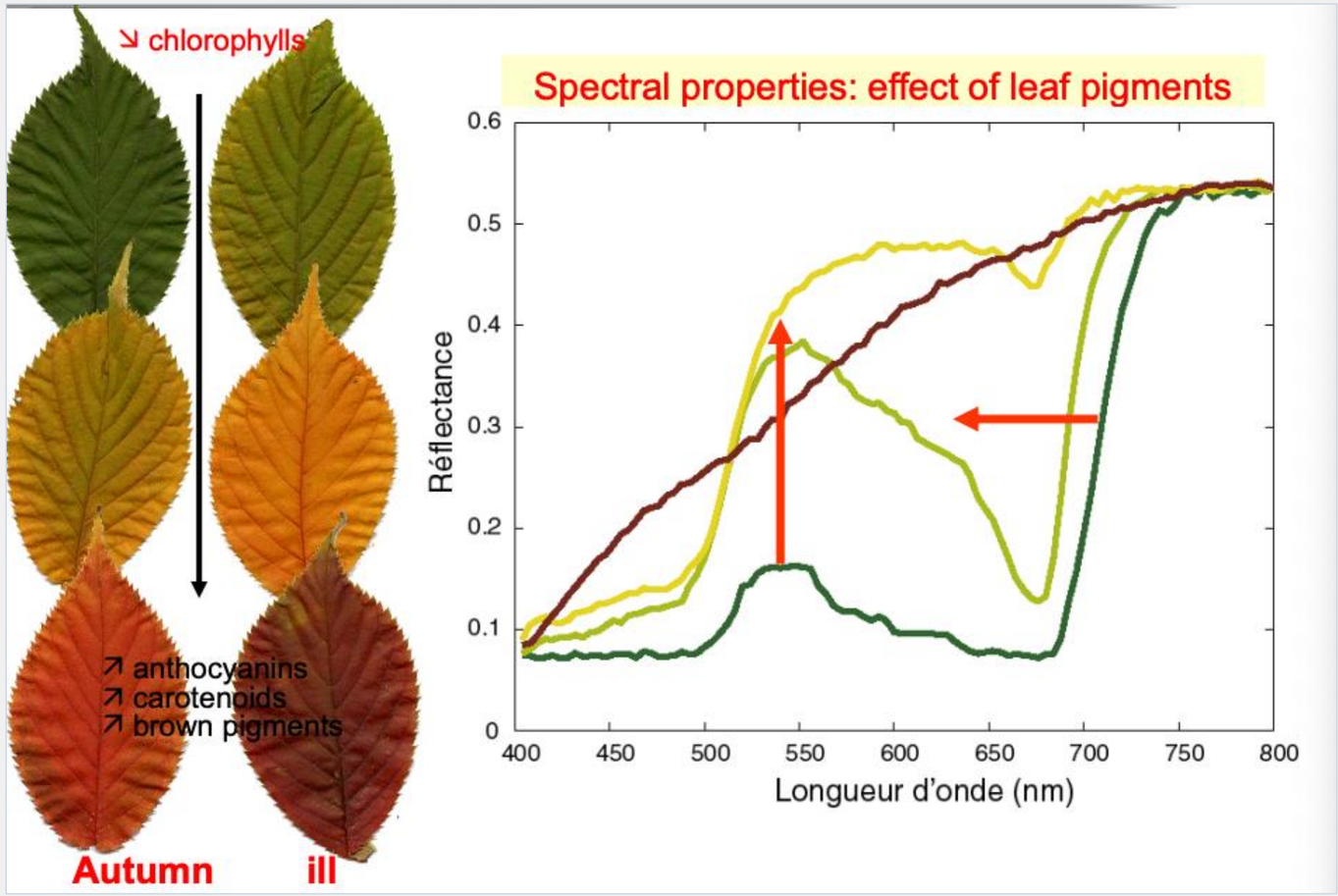
Leaf reflectance with the absorption features labelled

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Signatures change with condition

not a fixed value

- **Pigments change the curve.** As chlorophyll drops in autumn, the visible part of the spectrum rises.
- **Stress shows up early.** Often before a change is visible to the eye, which is the value for agriculture.
- **So always record the context.** Season, growth stage, moisture and sun angle all shift the curve.
- **Compare like with like.** A signature from one date and condition may not match another.



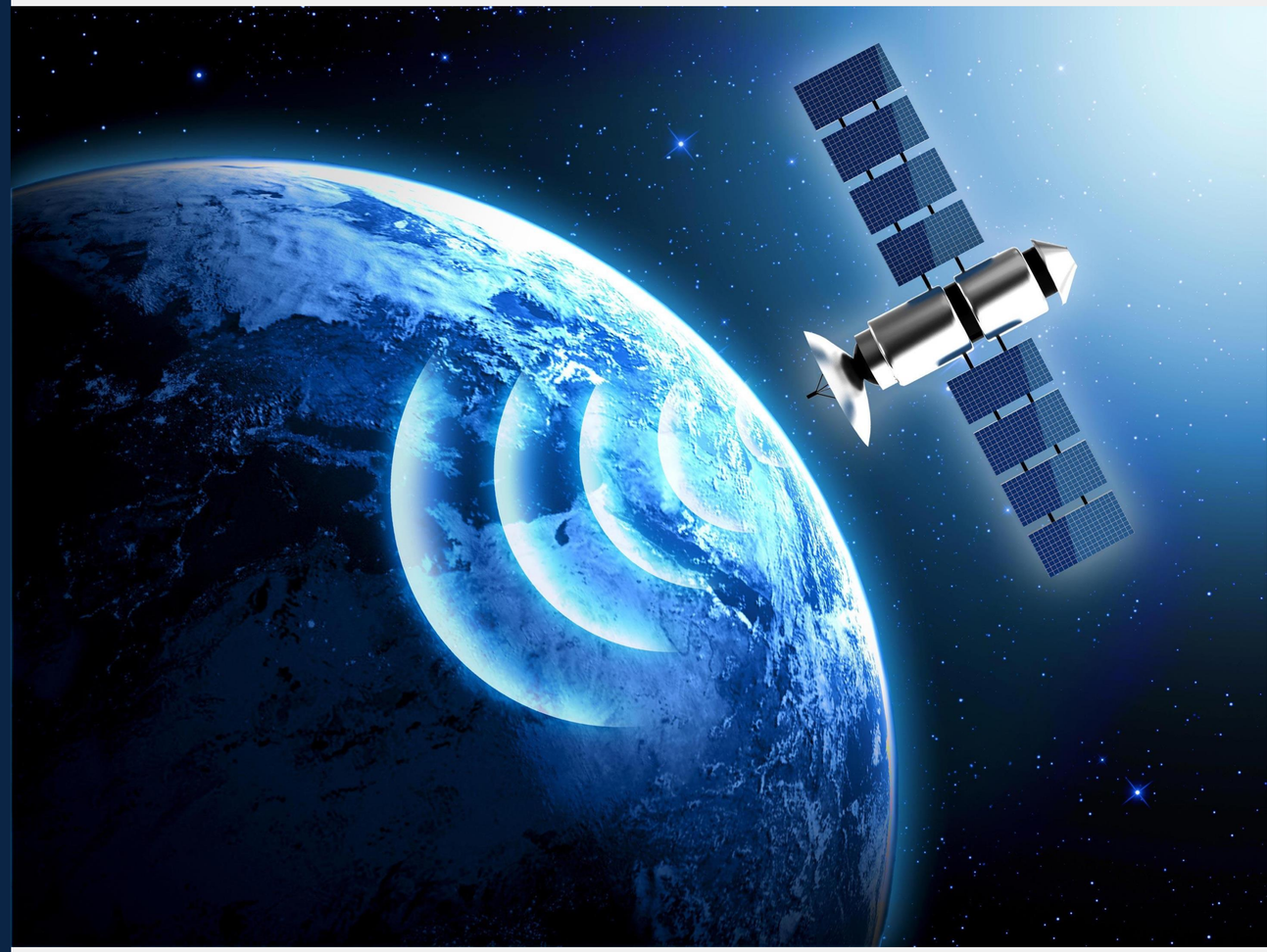
The same leaves in different states, and the effect on reflectance

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03

Data acquisition



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Field spectrometers

ground truth

- **Measurements on the ground.** Used to validate what the airborne or satellite sensor recorded.
- **Very high spectral detail.** Narrow bands, measured close to the target, with known illumination.
- **Builds spectral libraries.** The reference curves that later identify materials in an image.
- **Used across industries.** Agriculture, forestry, geology and materials inspection.



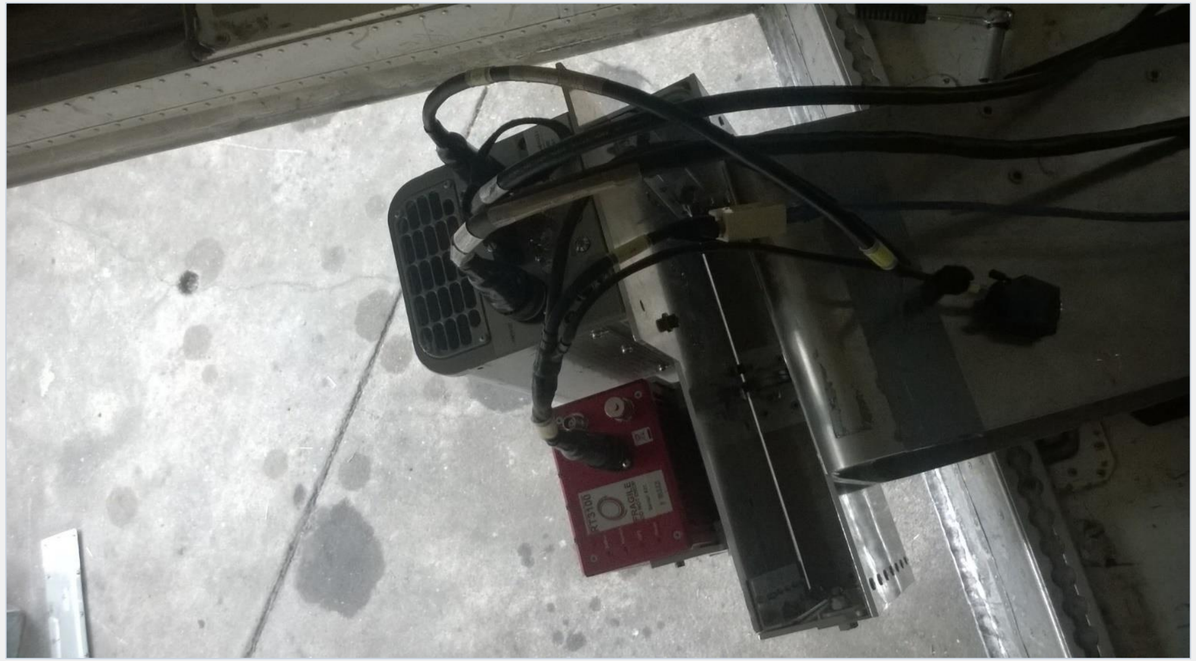
A backpack field spectrometer collecting reference spectra

Figure from the original UASA hyperspectral imaging course material

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Platforms · aircraft and drones

how we fly it



A hyperspectral sensor mounted in the aircraft floor



Survey aircraft being prepared for a campaign

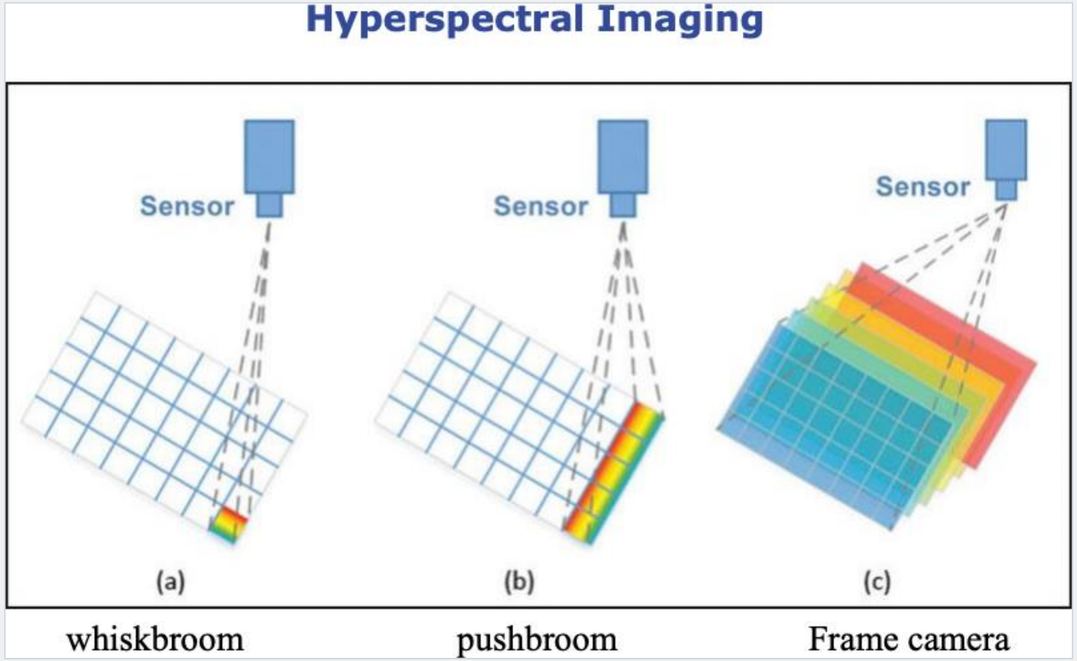
- **Aircraft.** Steady, reliable and able to cover a large area in one flight, but costly to mobilise.
- **Drones.** Cheap and flexible for small areas, though hyperspectral sensors are heavy for them.
- **Satellites.** Repeatable and global, with fewer hyperspectral missions and lower revisit.

Figure from the original UASA hyperspectral imaging course material

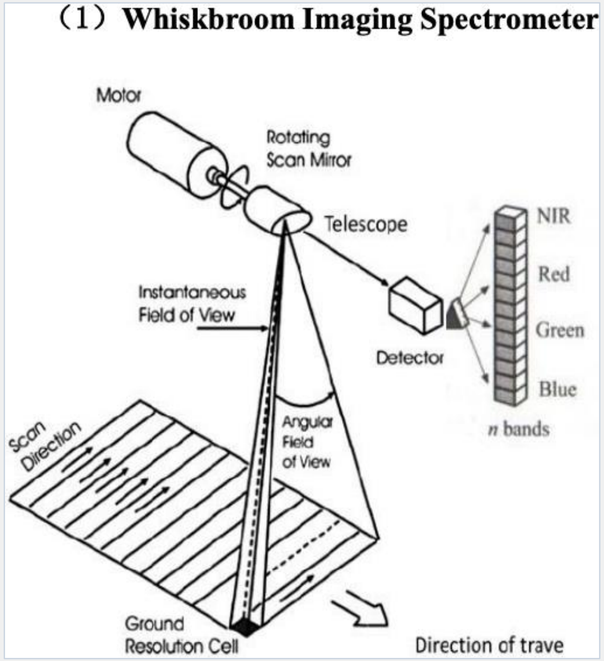
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Three ways to scan a scene

acquisition modes



Whiskbroom, pushbroom and frame camera



A whiskbroom scanner sweeps a mirror across the flight path

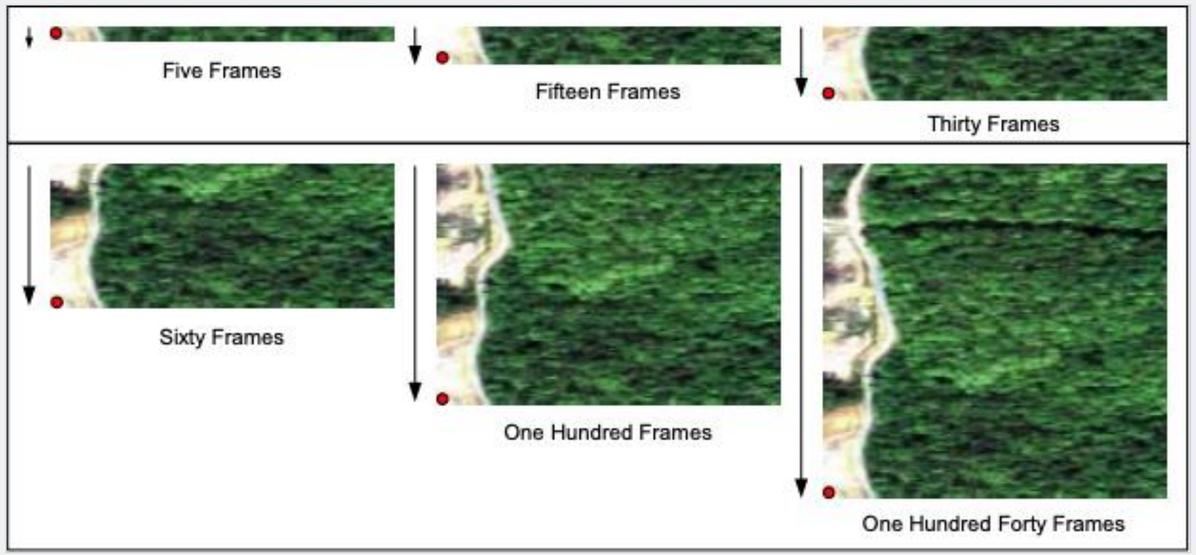
- **Whiskbroom.** A moving mirror sweeps point by point across the track. Mechanically complex.
- **Pushbroom.** A whole line of detectors is read at once as the platform flies forward. Most common today.
- **Frame camera.** Captures a full image at one wavelength at a time, tuning through the bands.

Figure from the original UASA hyperspectral imaging course material

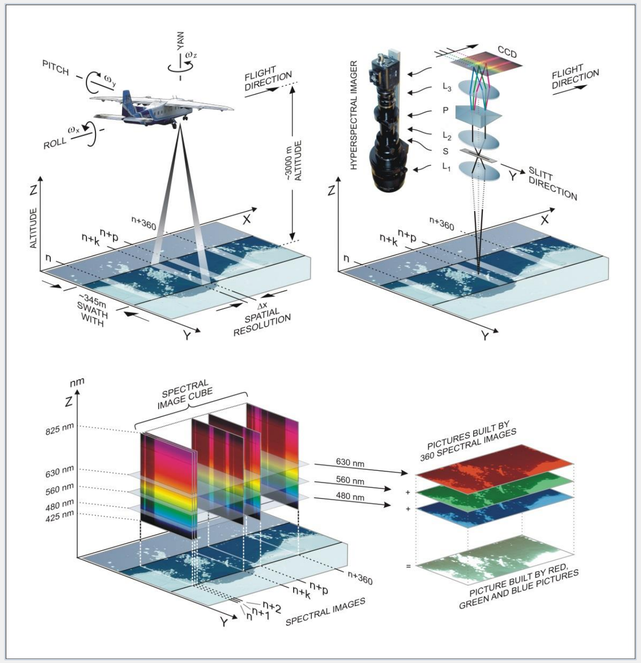
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The image is built line by line

pushbroom in practice



The scene grows frame by frame as the aircraft advances



Each line carries a full spectrum for every pixel

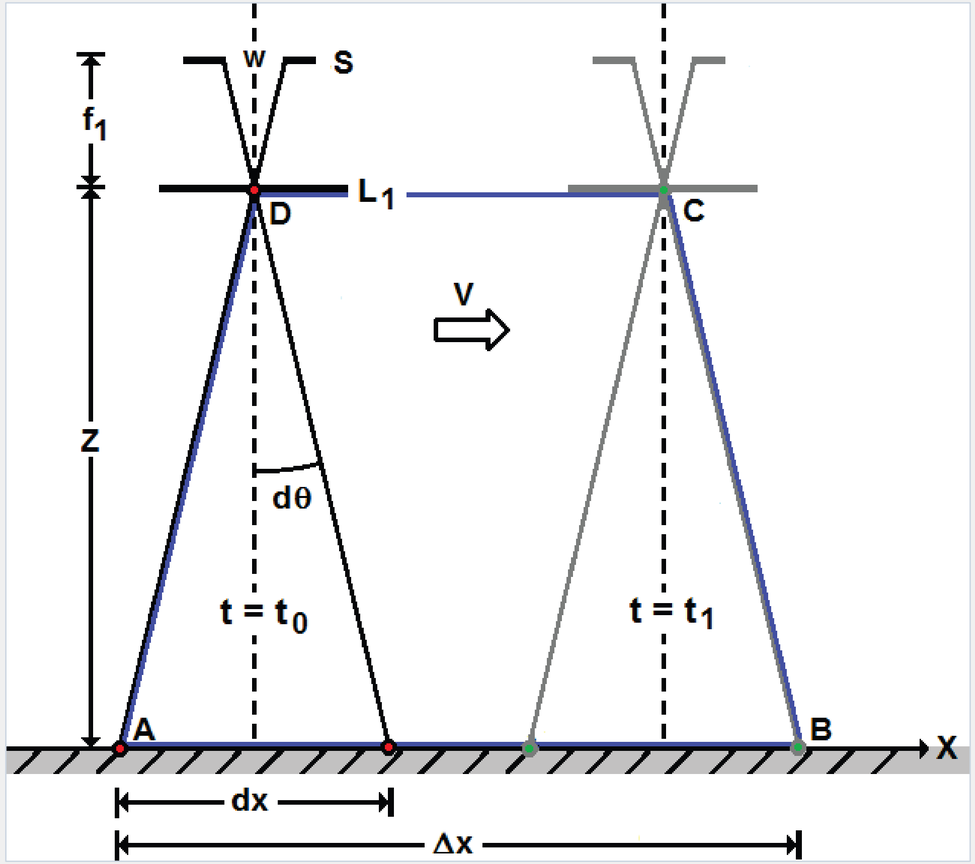
- **The frame rate sets the pace.** At 60 hertz a new line of pixels is captured every sixtieth of a second.
- **Speed and height must match the frame rate.** Too fast and the lines stretch, too slow and they overlap.
- **Position is recorded for every line.** GPS and an inertial unit log where the sensor was and how it was tilted.

Figure from the original UASA hyperspectral imaging course material

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What sets the pixel size

geometry



Flying height, field of view and speed decide the ground sampling

Flying height Z	Pixel size dx	Smear Dx	Swath width
100 m	0.16	0.36	18.75
300 m	0.47	0.67	56.25
500 m	0.78	0.98	93.75
1000 m	1.56	1.80	187.50

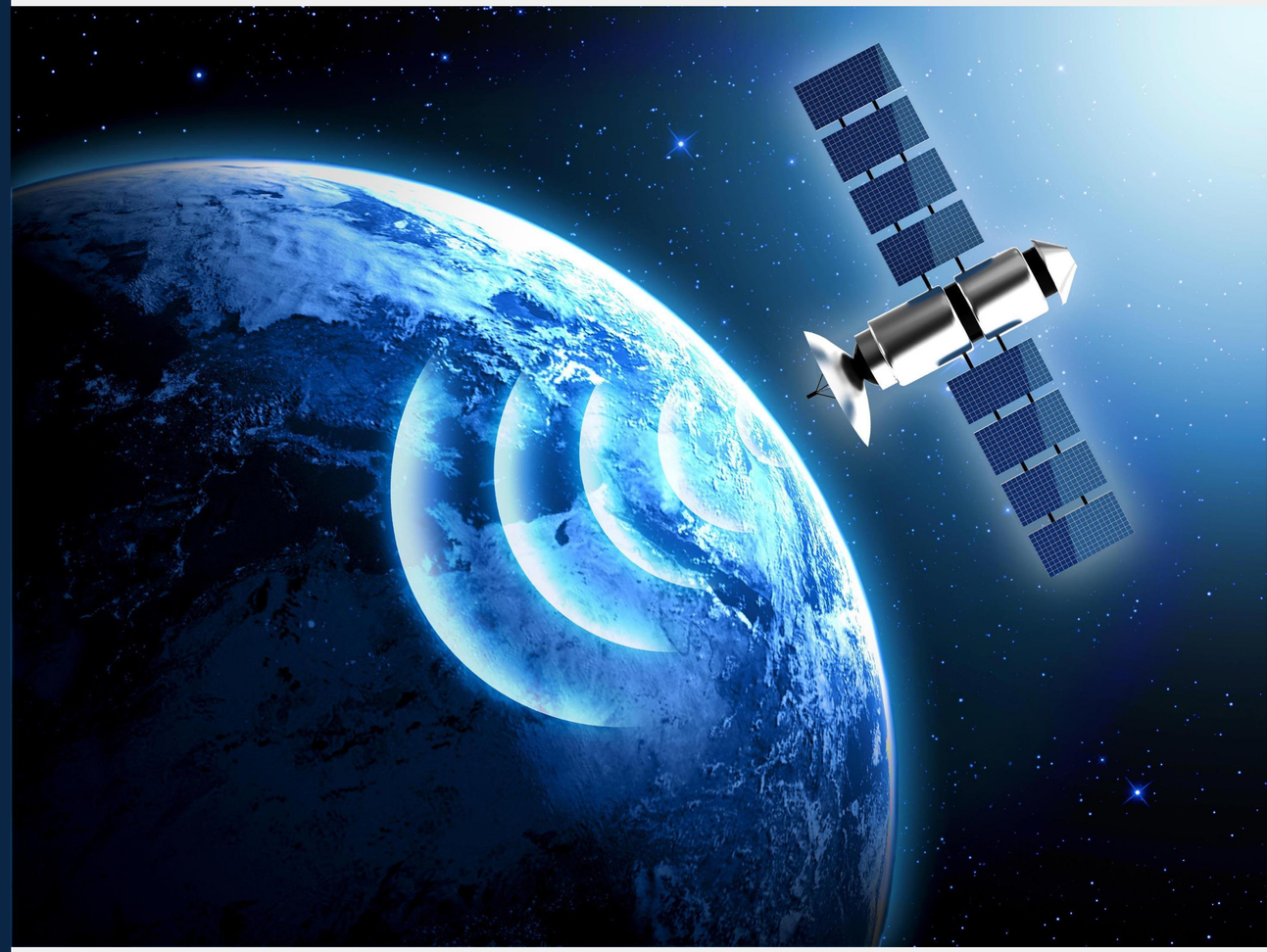
- **Fly lower for detail, higher for coverage.** Halving the height halves the pixel and the swath.
- **Smear is movement during one exposure.** It is always a little larger than the pixel itself.

Example calculation from the original course. Integration time 20 ms, 25 frames per second, ground speed 10 m/s. All values in metres.

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04

Processing HSI data



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The processing chain

raw data to answers

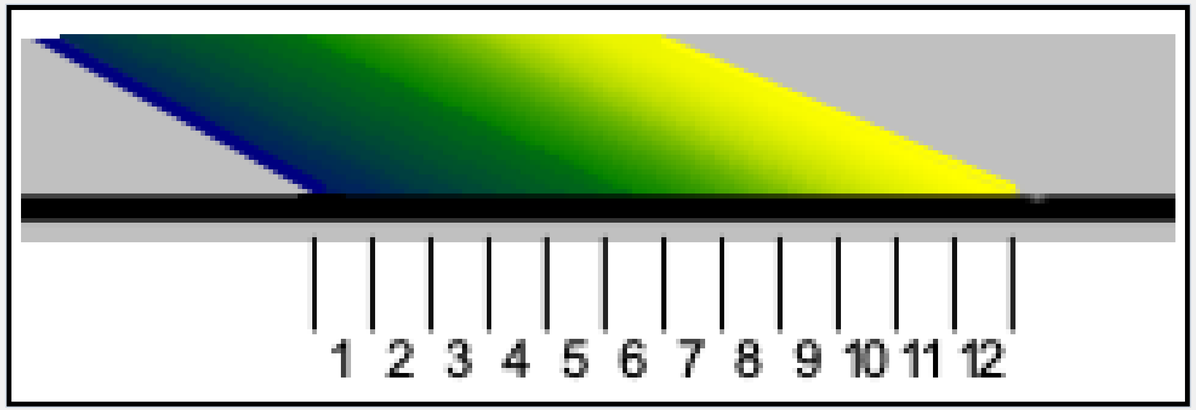
1	2	3	4	5
Radiometric calibration	**Binning**	**Atmospheric correction**	**Geometric correction**	**Analysis**
Give every channel its wavelength, and turn raw numbers into radiance.	Combine bands or neighbouring pixels to cut noise and size.	Remove the air, to get surface reflectance.	Put every pixel in its right place on the map.	Classify, detect and measure what is in the scene.

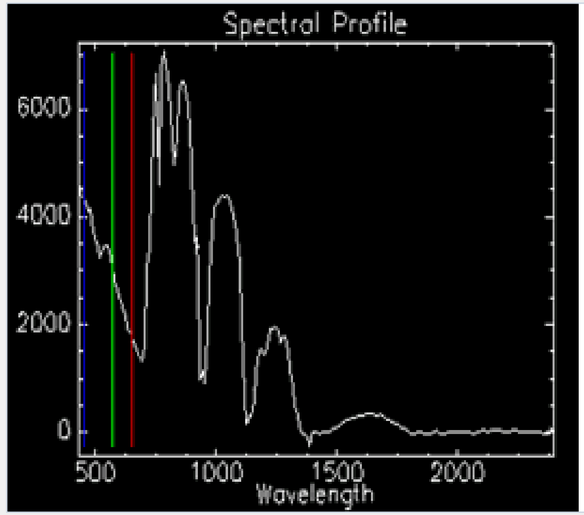
Each step assumes the one before it was done properly. Skip a step and the error travels all the way to your map.

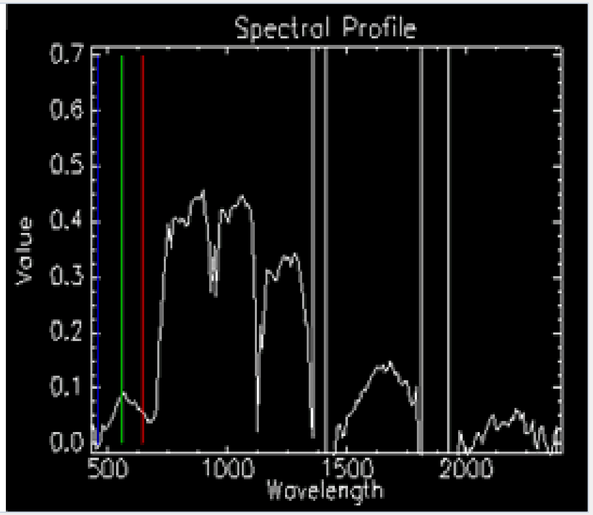
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Radiometric calibration

numbers to radiance







Each detector channel is assigned its wavelength

The same pixel as raw values, and as calibrated reflectance

- **Raw digital numbers mean nothing on their own.** They depend on the sensor, its gain and its temperature.
- **Dark current is removed first.** The signal the detector records with no light at all.
- **The result is a physical quantity.** Radiance, which can be compared between flights and sensors.

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Binning · trading detail for a cleaner signal

spectral and spatial

Spectral binning

Neighbouring bands are added together.

Fewer, wider bands with a stronger signal and less noise.

You lose some ability to separate materials.

Spatial binning

Neighbouring pixels are added together.

Bigger pixels, a stronger signal and faster processing.

You lose spatial detail, so small targets disappear.

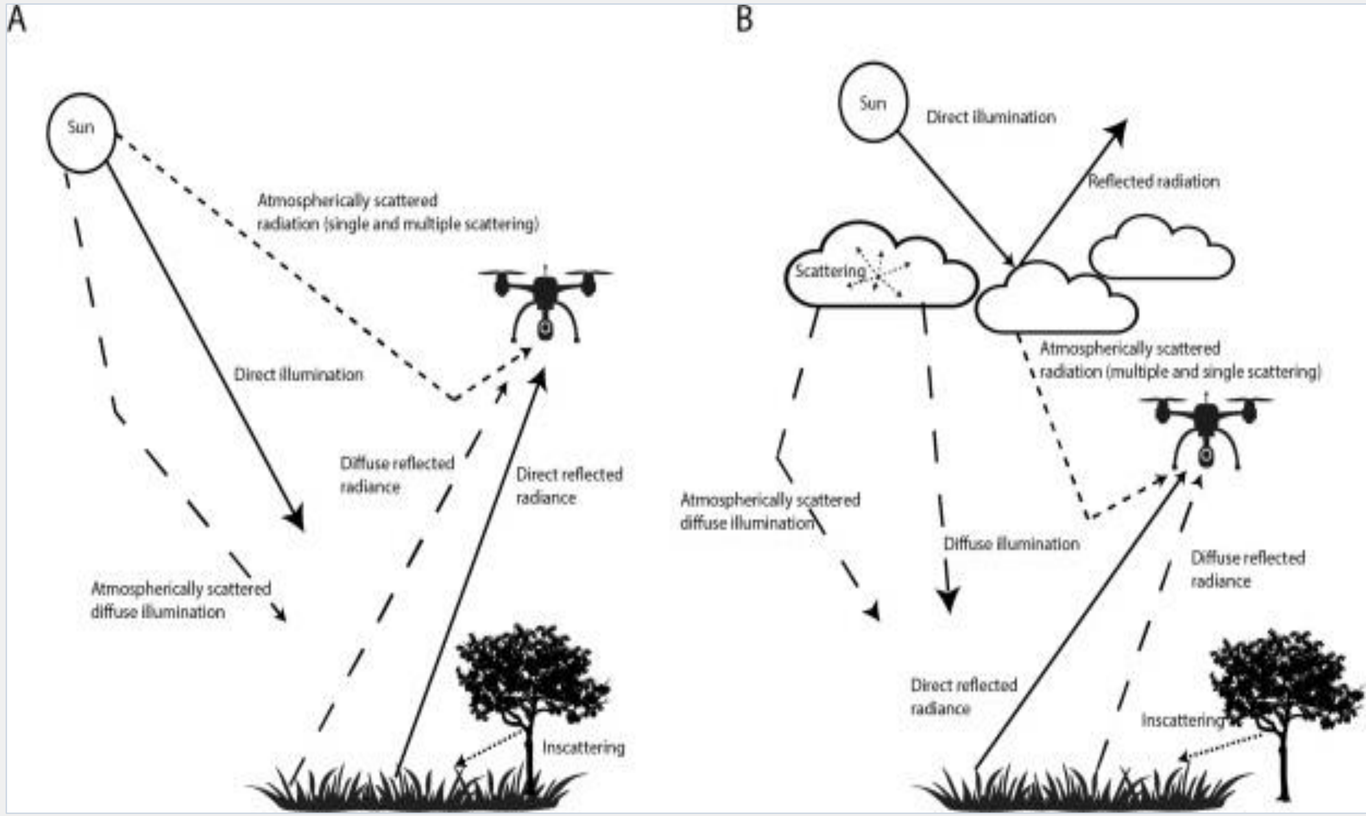
- **Hyperspectral data is big.** Hundreds of bands over a whole flight line add up quickly.
- **Binning is a deliberate choice.** Decide what you are willing to lose before you start, not afterwards.
- **Noise falls as the signal is pooled.** Which matters most in dark scenes, such as water.

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Atmospheric correction

removing the air

- **Not all light comes from the target.** Some was scattered by the air and never touched the ground, and some bounced off a neighbour.
- **Three paths reach the sensor.** Direct, single or multiply scattered, and diffuse light.
- **Water vapour blocks whole regions.** Those bands carry no usable surface signal and are normally discarded.
- **The goal is surface reflectance.** Only then can two dates, or two sensors, be compared honestly.

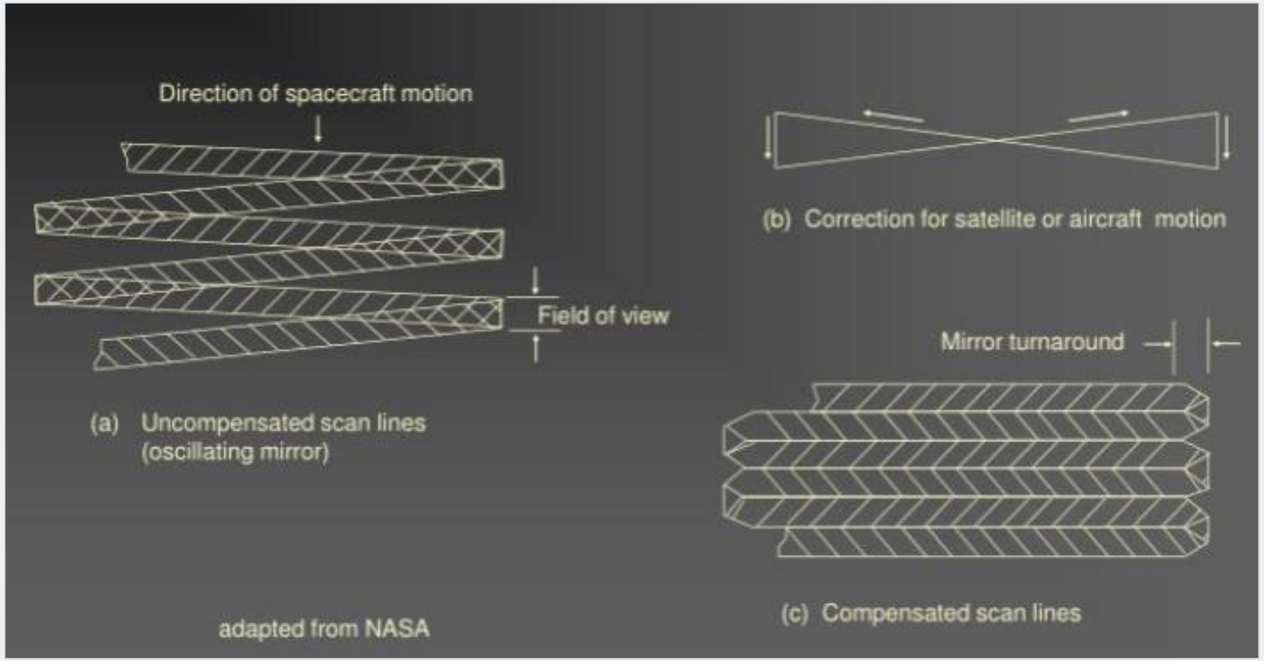


Light paths between the sun, the target and the sensor, in clear and cloudy air

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Geometric correction

putting pixels in place



Uncompensated scan lines, and the same lines corrected

- **The platform never flies perfectly.** Roll, pitch, yaw and gusts all bend the recorded scene.
- **Terrain adds its own distortion.** Hills shift pixels sideways unless a height model is used.
- **Orthorectification fixes both.** The output can then be overlaid on maps and on other datasets.

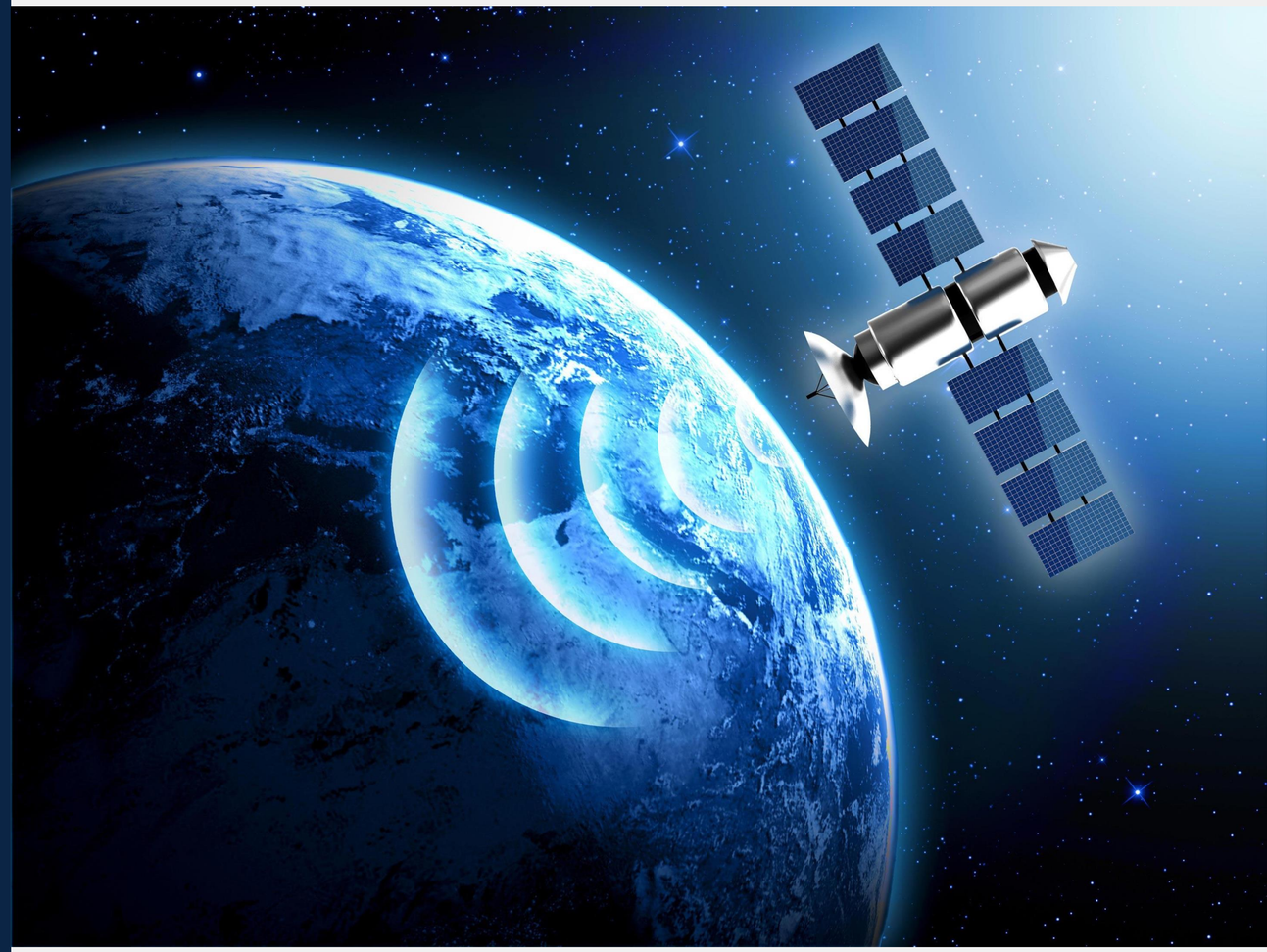


A flight strip after orthorectification

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05

Challenges



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Light and weather decide your data

illumination

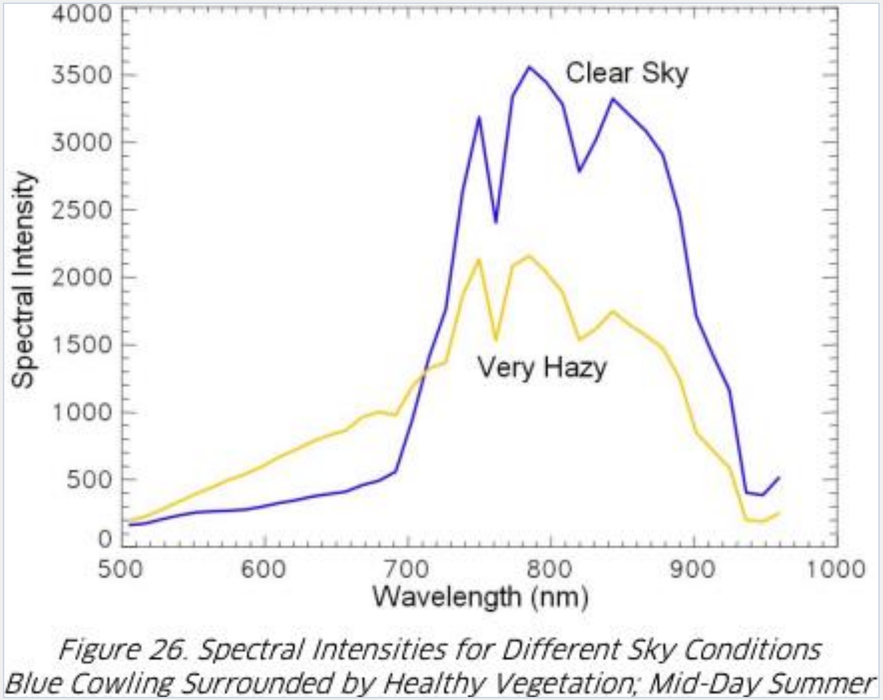
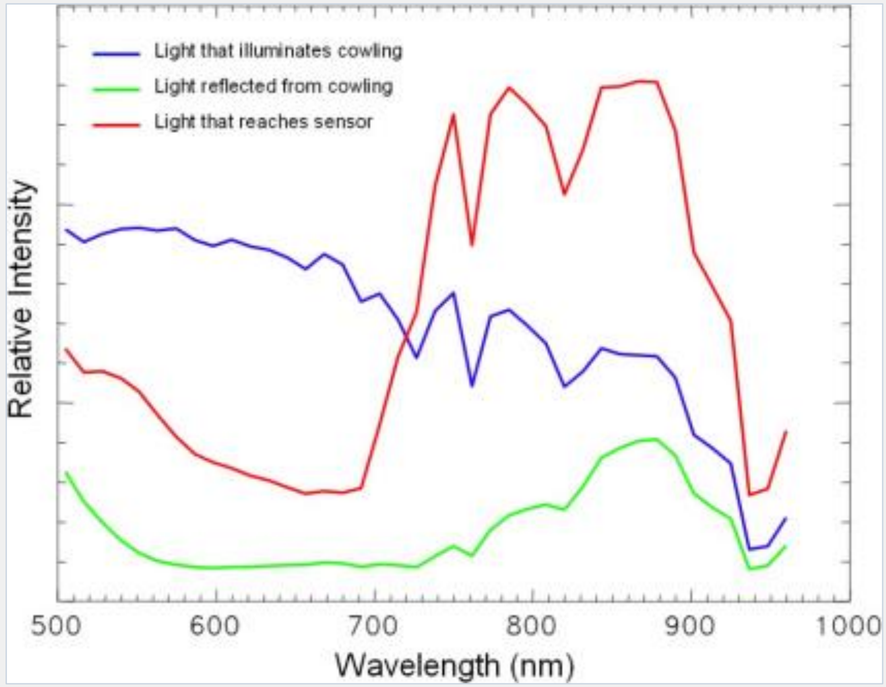


Figure 26. Spectral Intensities for Different Sky Conditions Blue Cowling Surrounded by Healthy Vegetation; Mid-Day Summer

The same target under clear and very hazy skies



Illuminating light, reflected light, and what actually reaches the sensor

- **Clouds and haze cost signal.** A hazy sky roughly halved the measured intensity in this example.
- **Sun angle matters.** A low sun means long shadows and weaker illumination.
- **Plan the flight window.** Clear sky, high sun and stable conditions, ideally near solar noon.

Figure from the original UASA hyperspectral imaging course material

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Movement, target and data size

the other challenges

- **Vibration and attitude.** Aircraft motion shifts lines, and a mismatch between the sensor and the position unit shows up as distortion.
- **Terrain relief.** Rugged ground distorts geometry and casts shadows that hide the target.
- **Target condition.** Leaf on or leaf off, wet or dry, the state of the target must match your question.
- **Data volume.** Hundreds of bands per pixel need storage, computing power and careful handling.
- **Skilled interpretation.** The instrument is only as good as the person reading the spectra.

Figure 3



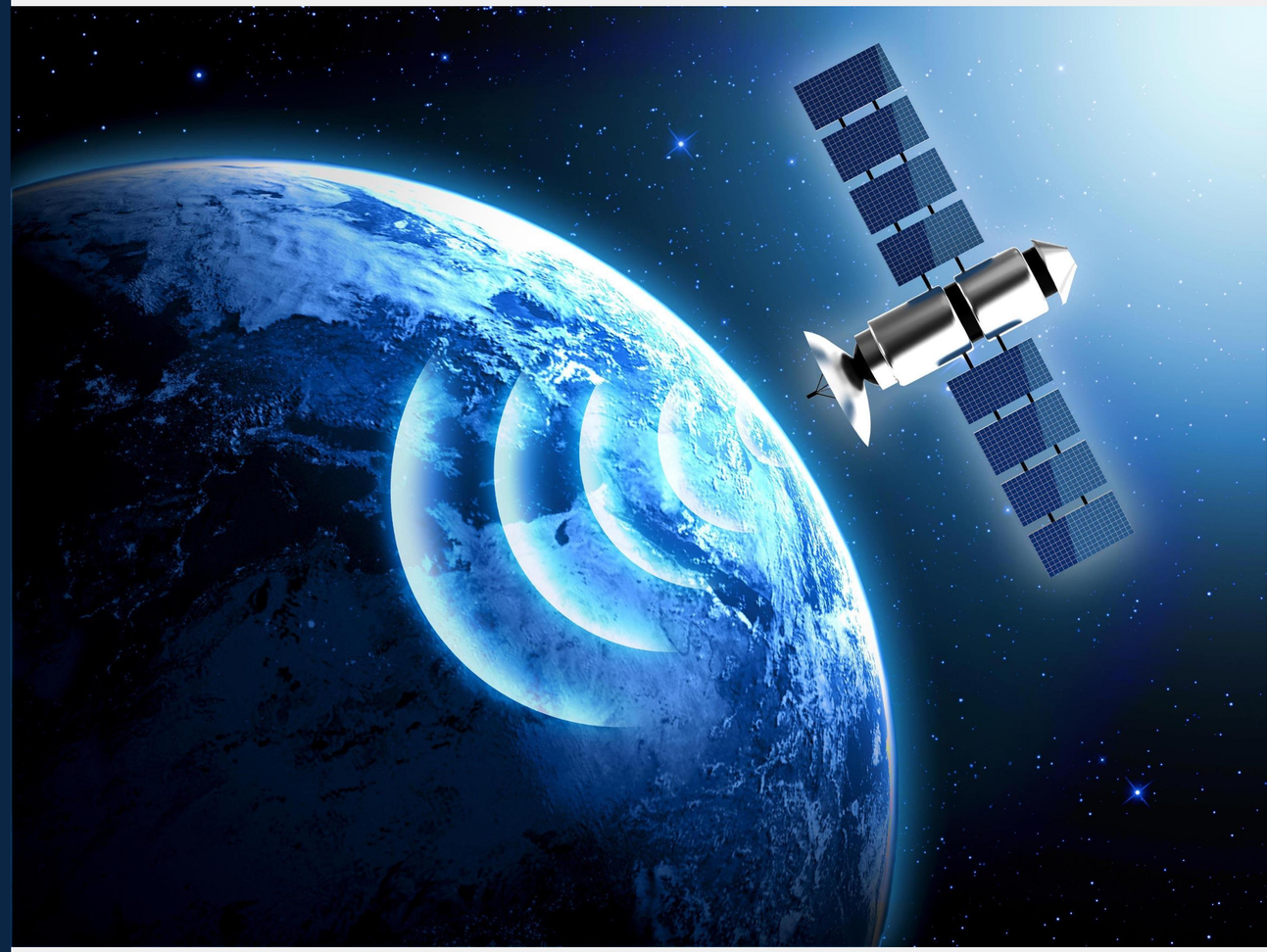
Geometric distortion along an uncorrected flight strip

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06

Hyperspectral sensors



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Sensors flying today

missions

Instrument	TianGong-1	PRISMA	HISUI	EnMAP	SHALOM	HypXIM
Organization	Chinese Academy of Science and Physics	Italian Space Agency (ASI)	Japanese ministry of Economy, Trade, and Industry	German GFZ-DLR	Italy-Israel Space agencies (ASI-ISA)	France Space Agency (CNES)
Date Range	2011-2013	2020-Present	2021	2021	2022	2021/2022
Spectral Range	400-2500 nm	400-2500 nm	400-2500 nm	420-2450 nm	400-2500 nm	400-2500 nm
Spectral Bands	128	249	185	244	275	210
Spatial Resolution	10-20 m	30 m	30 m	30 m	10 m	10 m
Objective	Land imaging in China	Natural resources and atmosphere	Energy, vegetation monitoring	Earth observation	Land and ocean observation	Soil, urban, coastal applications

Instrument comparison

Mission	Who	Status
PRISMA	ASI, Italy	in orbit since 2019
EnMAP	DLR, Germany	in orbit since 2022
EMIT	NASA, on the ISS	operating since 2022
FLEX	ESA	launch set for September 2026
CHIME	ESA Copernicus	in development

- **Most hyperspectral data is still airborne.** Satellite missions are growing but revisit is low.
- **Check the status before teaching.** Launch dates move, and new cube-sat constellations appear each year.

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The list keeps growing

- **Cube-sats changed the picture.** Small satellites from several companies now offer hyperspectral data on demand.
- **Sensors are getting lighter.** Which brings hyperspectral imaging within reach of drones.
- **Public missions anchor the field.** They provide calibrated, free data that everyone can validate against.
- **Plan for a mixed world.** Combine campaign flights for detail with satellites for repeat coverage.

what is coming

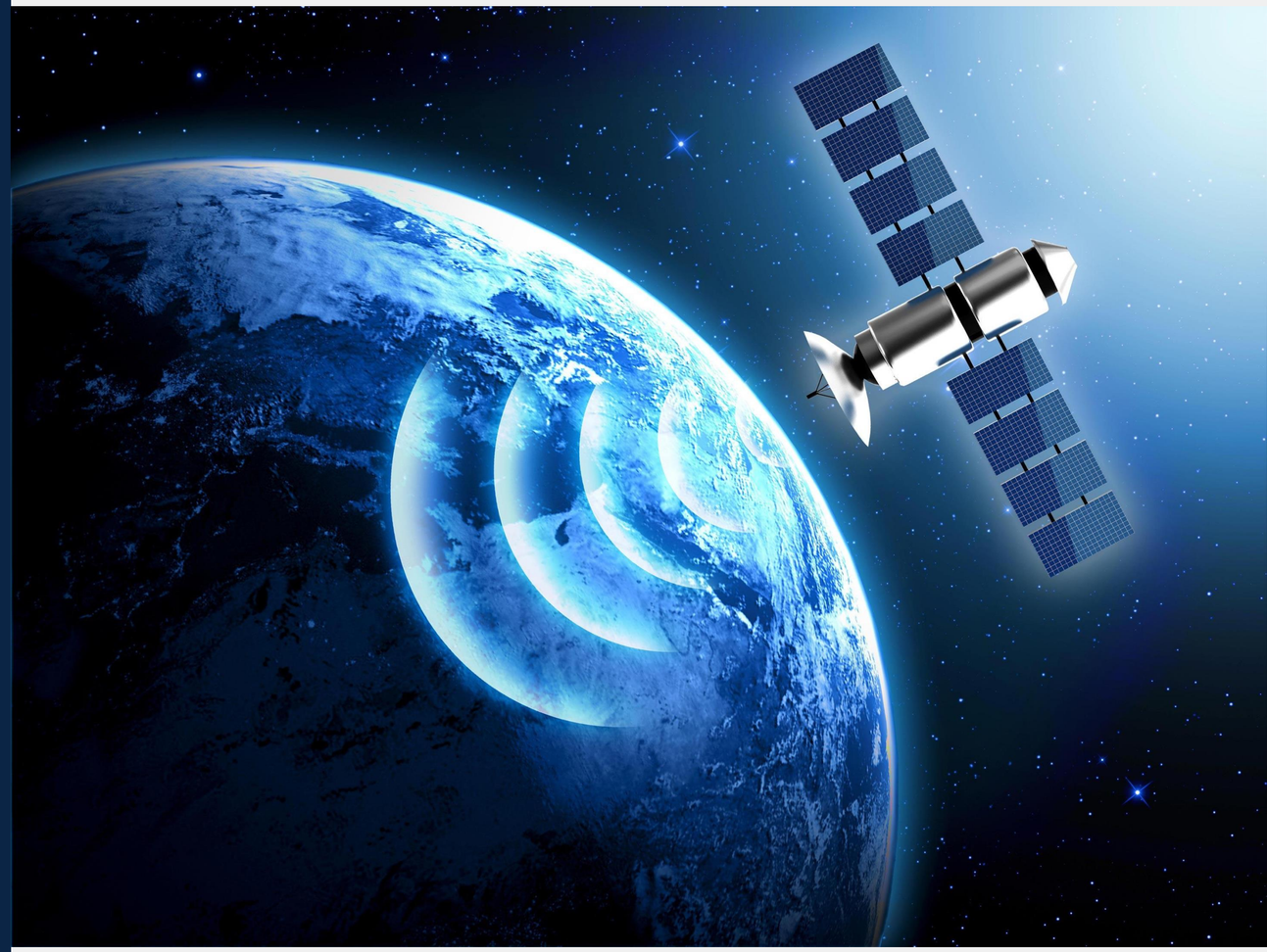
Launch date	Name	Provider	Platform	Constellation size now / planned	Wavelengths in nm	Spectral res. in nm	Spatial res. in m
N/A	AVIRIS (classic)	NASA/JPL	Airborne		380-2510	10	Varies
N/A	AVIRIS-NG	NASA/JPL	Airborne		380-2510	5	Varies
2018	DESIS	DLR	ISS		400-1000	2.55	30
2019	PRISMA	ASI	Satellite		400-2500	10	30
2022	EnMAP (VNIR)	DLR	Satellite		420-1000	6.5	30
2022	EnMAP (SWIR)	DLR	Satellite		900-2450	10	30
2020-25	Satellogic Aleph-1	Satellogic	Satellite	38 / 200+	460-830	14-35	25
2022-23	PIXXEL	PIXXEL	Satellite	6			5
2023-24	Orbital Sidekick GHOSt	Orbital Sidekick	Satellite	3 / 6	400-2500	3-5	8.3
2023	Esper	Esper	Satellite	18	400-2000	6	6
2023	Carbon Mapper (Tanager)	Planet/JPL/CarbonMapper Coalition	Satellite	2		5	30
2023	Wyvern	Wyvern Space	Satellite	36			<5
2023	HySpec	HyspeqIQ	Satellite	12		105	5
2023	HyperSat	HyperSat	Satellite				6
2023	Kuva Hyperfield-1	Kuva Space	Satellite		450-1000		25
early 2024	PACE	NASA	Satellite	1	350-885	1.25 or 2.5	
2024	PIXXEL	PIXXEL	Satellite	30	400-2500		5
2027/28	SBG	NASA	Satellite			10	30
2029	CHIME (Sentinel 10)	ESA	Satellite		400-2500	10	
Unknown	BlackSky	BlackSky	Satellite	Unknown			

Hyperspectral missions and their launch dates, from the course material

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07

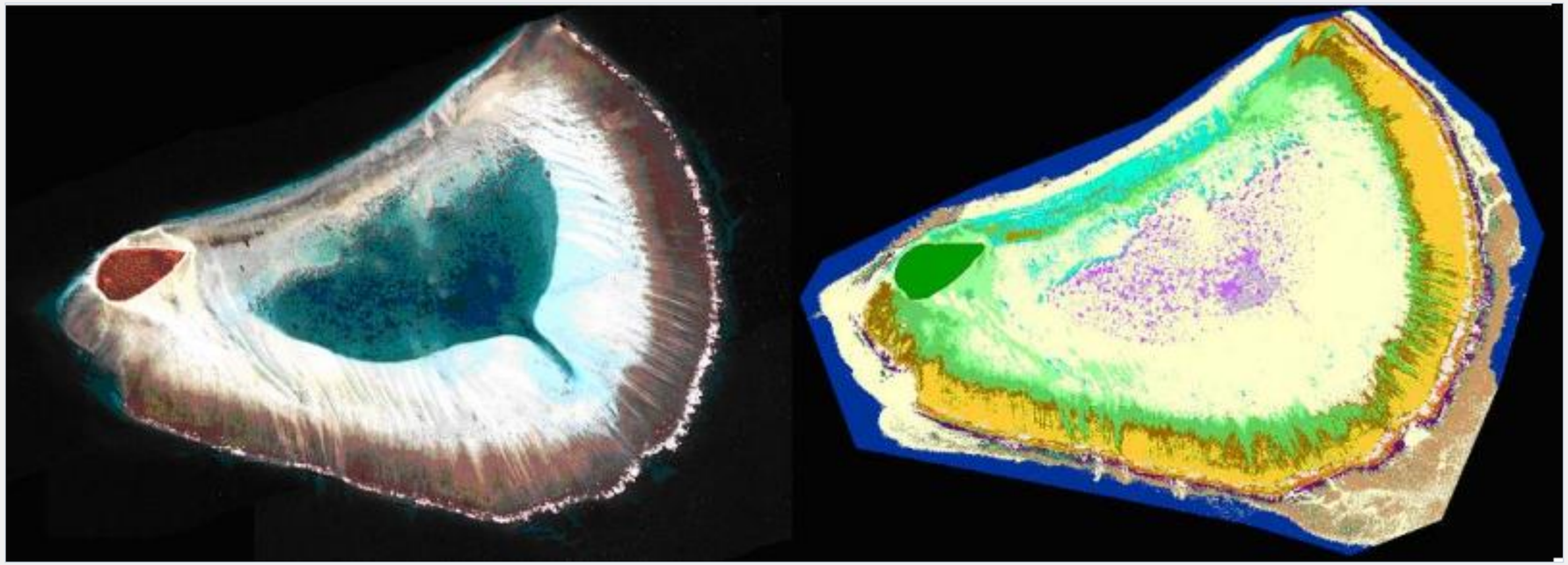
Applications



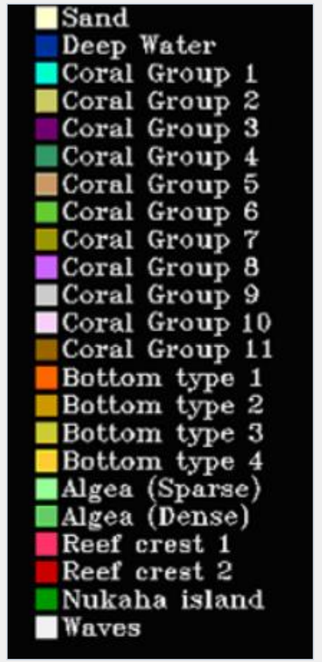
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Environment and coasts

applications



Hyperspectral data of Fordate island, Tanimbar, and the bottom-type map made from it



Classes separated in the map

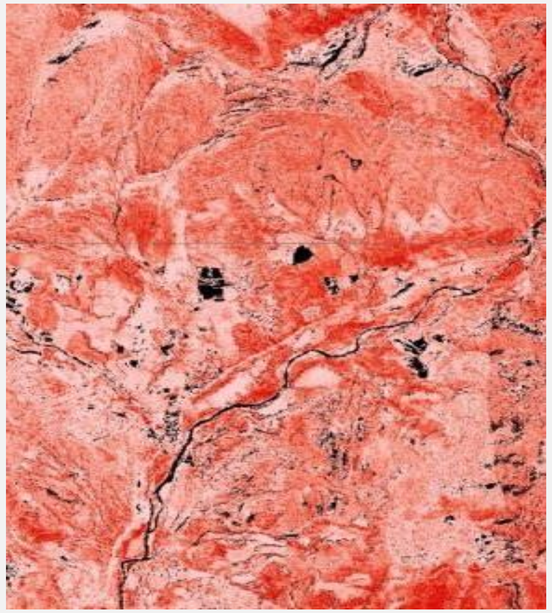
- **Shallow water bottom mapping.** Coral groups, algae, sand and reef crests separated by their spectra.
- **Geology and soils.** Mineral composition, soil texture and mine waste.
- **Vegetation and carbon.** Species mapping, plant stress, invasive species and carbon monitoring.

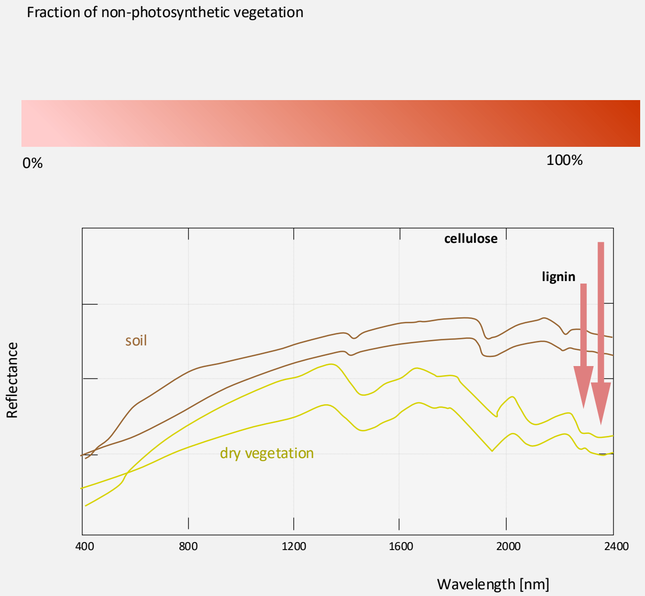
Image credit SEOS, via the original course material

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Mineral and geology

Dry vegetation Mapping

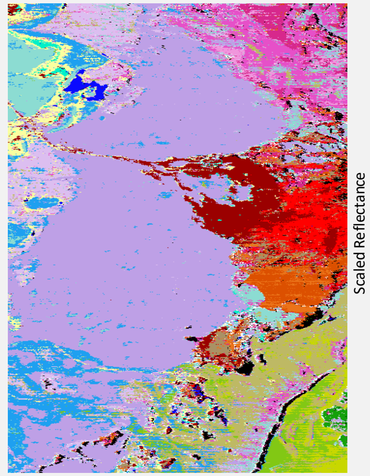


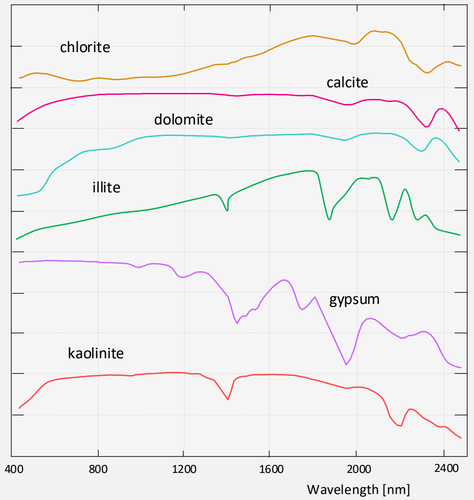


- **Geology and soils.** Mineral composition, soil texture and mine waste.
- Mine exploration
- Soil mineralogy, and texture
- Vegetation monitoring: plant stress, crop residues, plant composition

Image credit SEOS, via the original course material

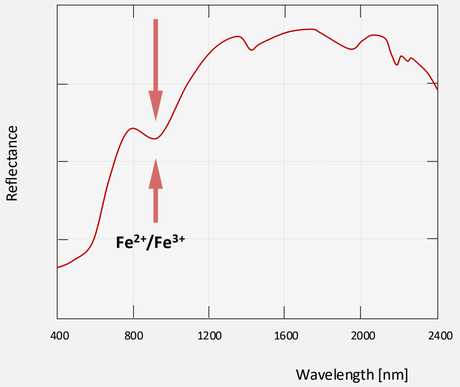
Mineral mapping





Soil mapping

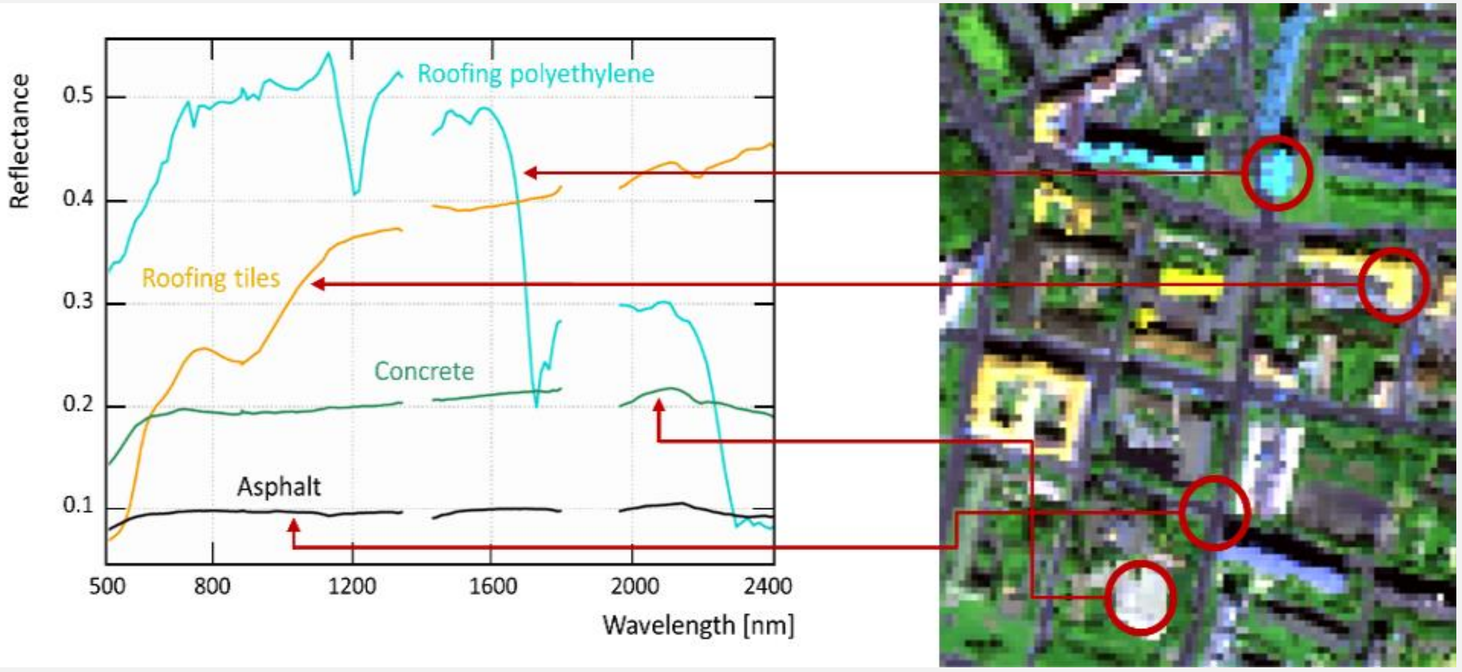




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Introduction to Hyperspectral Imaging

URBAN SURFACE MAPPING

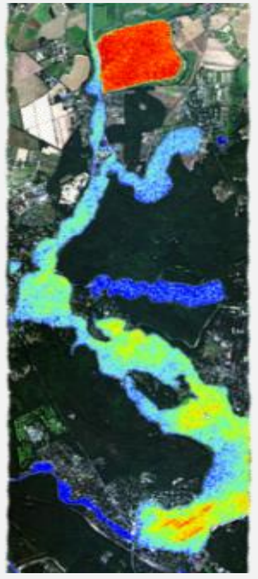


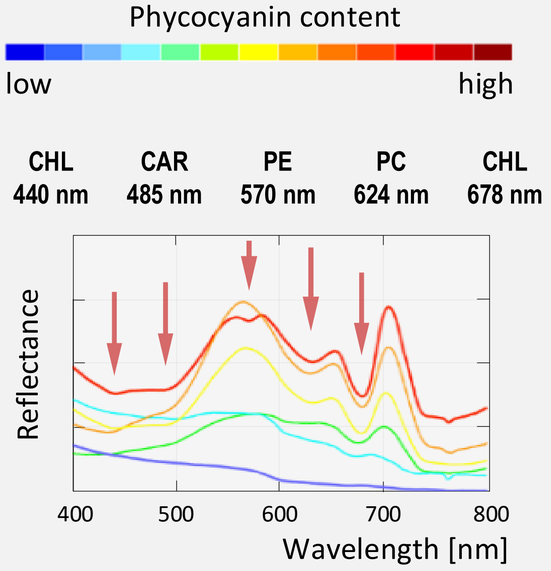
- Hazardous materials: oil, dust, mine waste, pollutants, plastics
- Inland and coastal waters: phytoplankton composition, (harmful) algae species
- Urban surface materials

Image credit SEOS, via the original course material

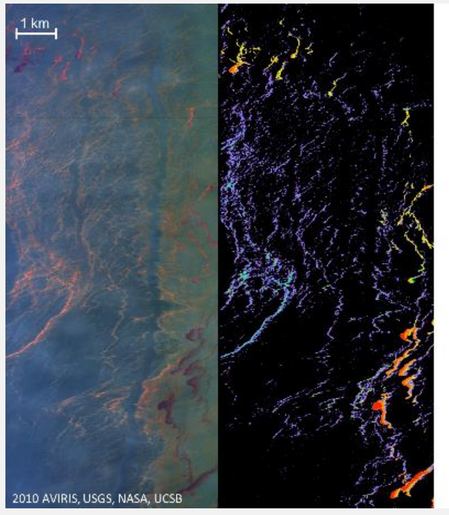
Classes separated in the map

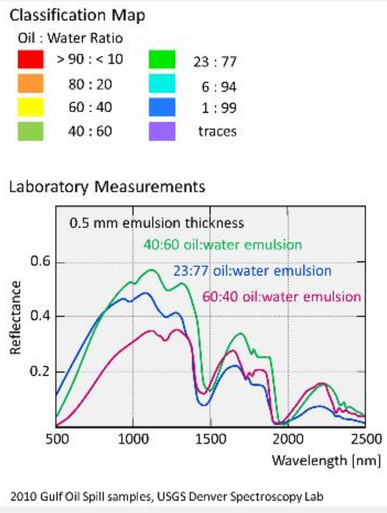
WATER QUALITY MAPPING





POLLUTION MAPPING



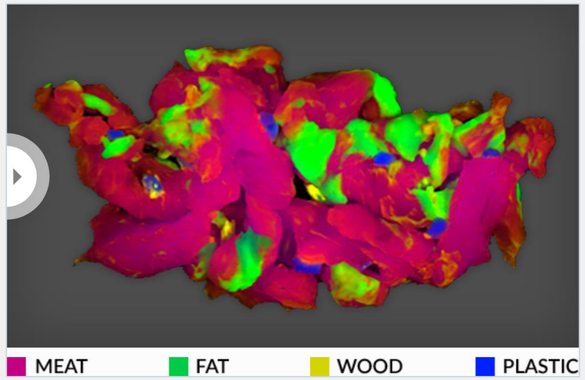


Food and packaging

industry



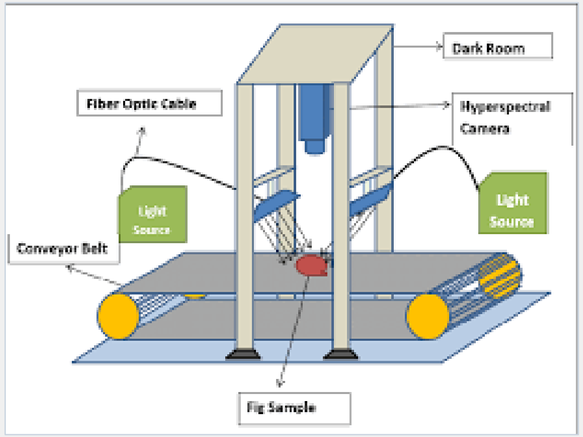
Meat trimmings as the camera sees them



The same tray classified into meat, fat, wood and plastic



Contamination found inside sealed packaging



A typical conveyor installation with halogen lamps

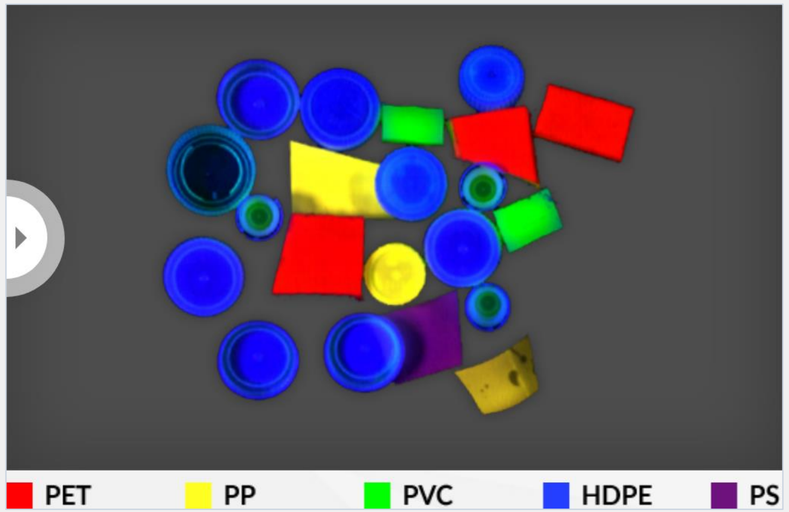
- **Chemistry, not colour.** Materials that look identical separate cleanly in the shortwave infrared, around 900 to 1700 nanometres.
- **Controlled light.** Indoor systems use halogen lamps, so illumination is stable and correction is simple.

Figure from the original UASA hyperspectral imaging course material

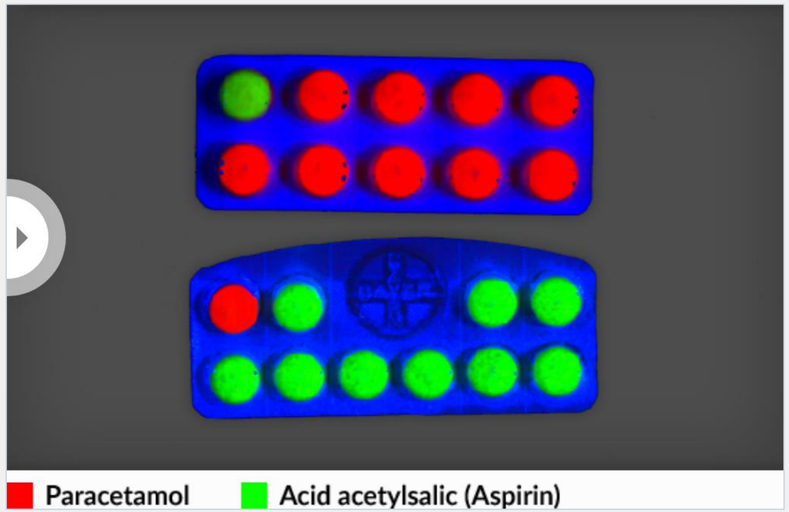
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Recycling, pharmaceuticals and mining

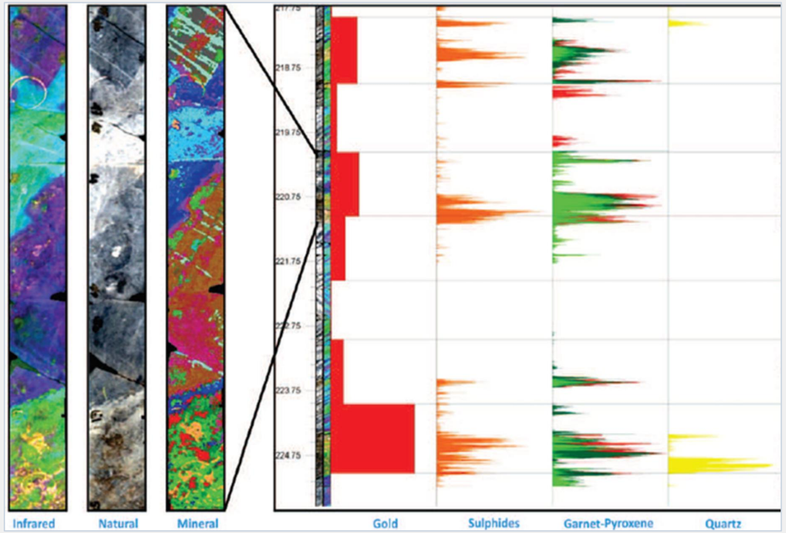
industry



Plastic types separated automatically for sorting



Two medicines that look alike, told apart by their spectra



Drill core scanned for minerals, with the mineral profile alongside

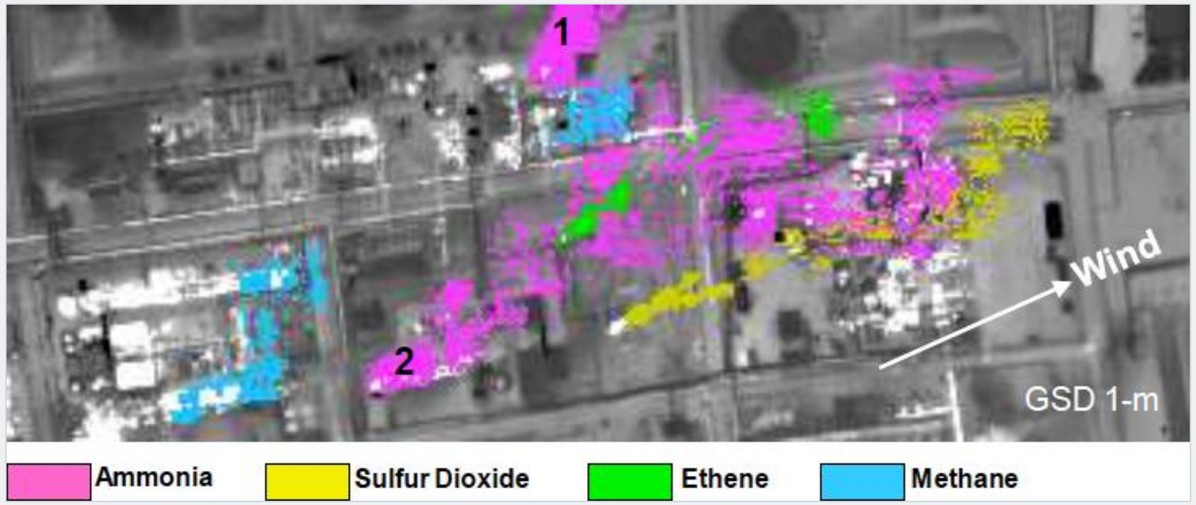
- **Sorting lines.** Plastics are identified by polymer type at conveyor speed.
- **Quality control.** Tablets, powders and packaging checked without opening or destroying them.
- **Core logging.** Whole boreholes scanned for mineralogy in hours instead of weeks.

Figure from the original UASA hyperspectral imaging course material

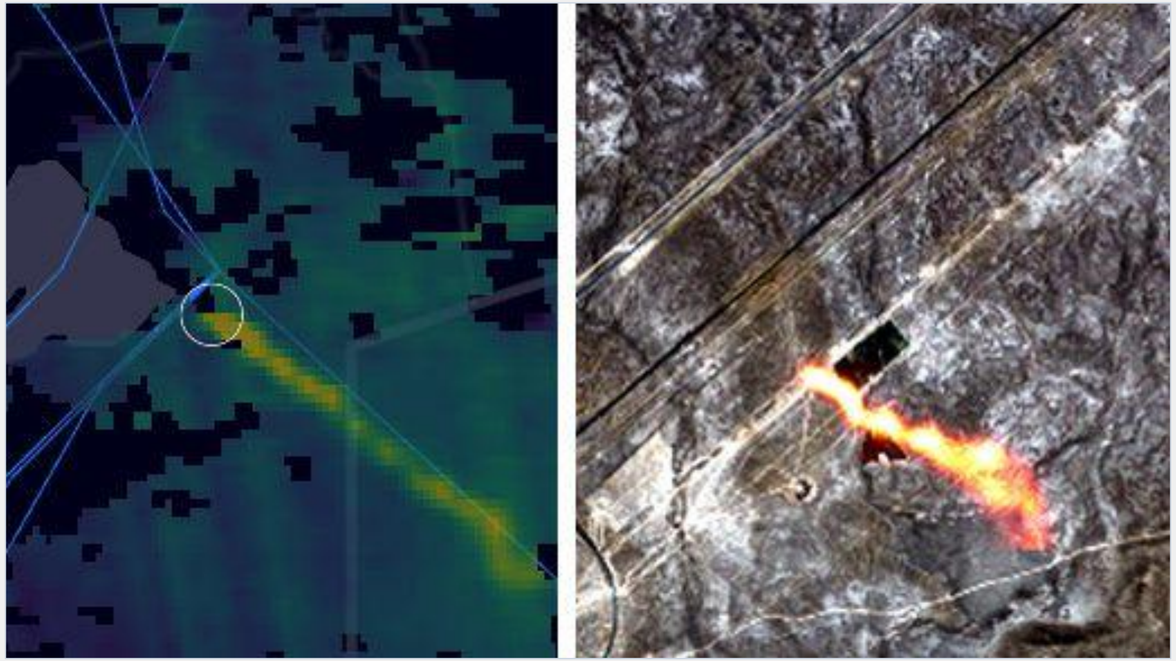
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Gas and leak detection

applications



Ammonia, sulphur dioxide, ethene and methane plumes mapped over an industrial site



A methane pipeline leak in Kazakhstan seen from orbit

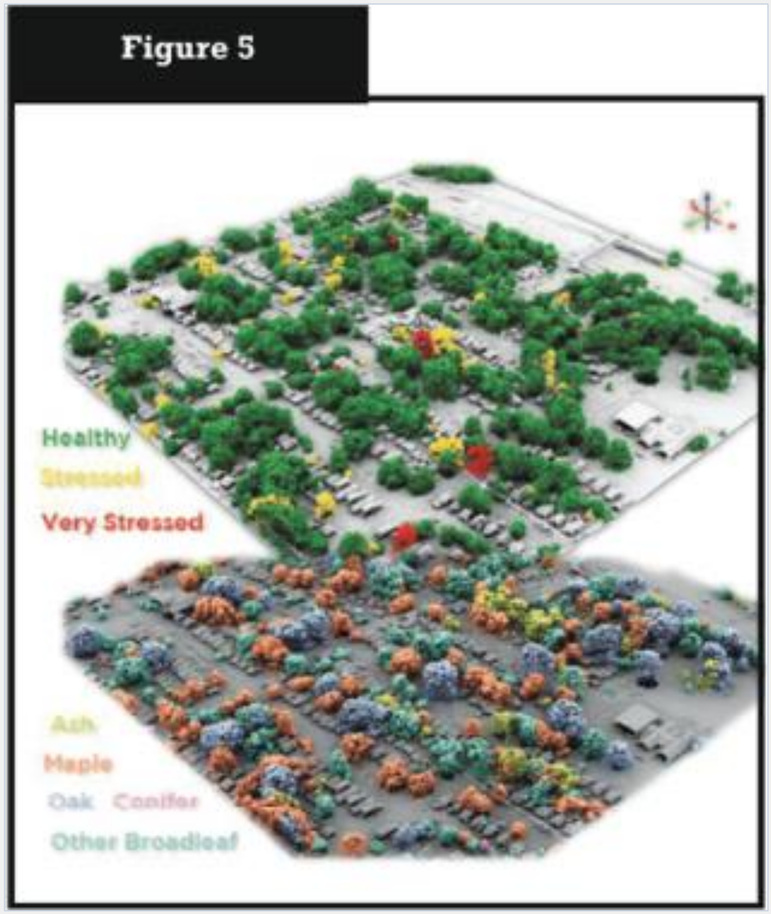
- **Gases have spectral fingerprints too.** Mostly in the mid and long-wave infrared, which needs a thermal sensor.
- **Aircraft find the leak, satellites find the region.** Ground and airborne systems localise, while satellites screen wide areas.
- **A growing market.** Emissions monitoring is driving a new generation of greenhouse-gas satellites.

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HSI and LiDAR together

data fusion

- **They measure different things.** LiDAR gives shape and height, hyperspectral gives material and condition.
- **Together they describe a forest.** Species and health from the spectrum, height and structure from the point cloud.
- **Flown on the same platform.** Sensors are often mounted together so the two datasets share one geometry.
- **Also fused with thermal.** Temperature adds moisture stress and gas detection to the picture.



Tree species and stress from hyperspectral data, over a LiDAR-derived structure

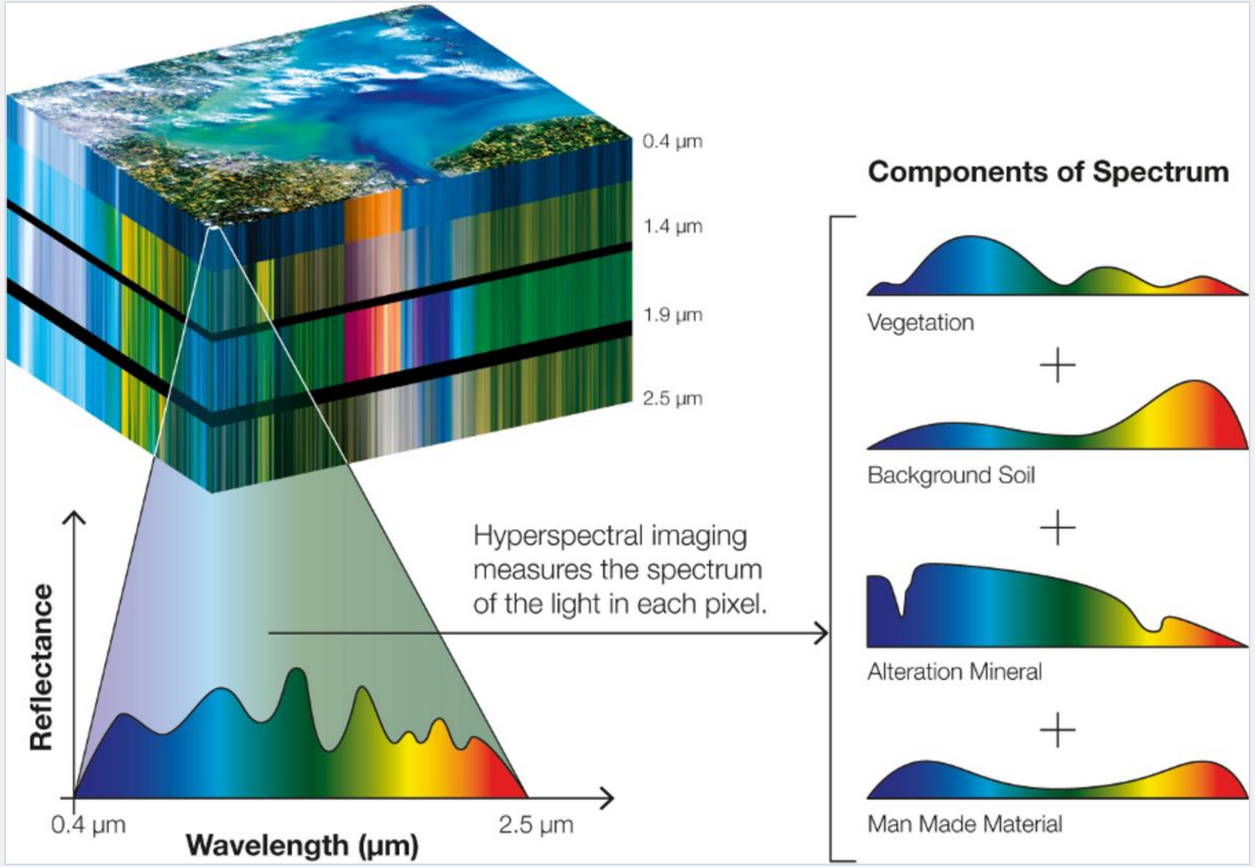
Figure from the original UASA hyperspectral imaging course material

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Five things to remember

summary

- **A spectrum is a measurement, not a picture.** Hundreds of narrow bands per pixel, which is how look-alike materials are told apart.
- **Physics decides what is possible.** Illumination, atmosphere and the target's condition set the limits before any software runs.
- **Order of processing matters.** Calibration, then atmosphere, then geometry. An error early on travels all the way to the map.
- **The data is large and complex.** Plan storage, binning and processing time from the start of the project.
- **The field is moving quickly.** Cube-sats, lighter sensors and fusion with LiDAR and thermal are widening what is practical.



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Thank you

Questions, and then we look at real hyperspectral data